

Decomposition Techniques for Social Epidemiology: Why and How

Danish Epidemiological Society Conference

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and Occupational Health

Social Epidemiologic Aims

Descriptive

Existence of social differences in health

Etiologic

Causes of social differences in health

Interventions

Policies and programs to address causes

Enough already?

Nevertheless, health inequalities are socially produced, and therefore, also potentially avoidable. However, effective political interventions require a scientific understanding of the causal mechanisms generating the strong and persistent correlations between social conditions and health outcomes.

How can we find out?

Eikemo and Øversveen (2019)

Scandinavian Journal of Public Health
Volume 47, Issue 6, August 2019, Pages 593-597
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<https://doi.org/10.1177/1403494819866416>



Editorial

Social Inequalities in health: Challenges, knowledge gaps, key debates and the need for new data

Terje A. Eikemo and Emil Øversveen

Decomposition: from description to explanation

What changed?

Explaining growing or shrinking gaps

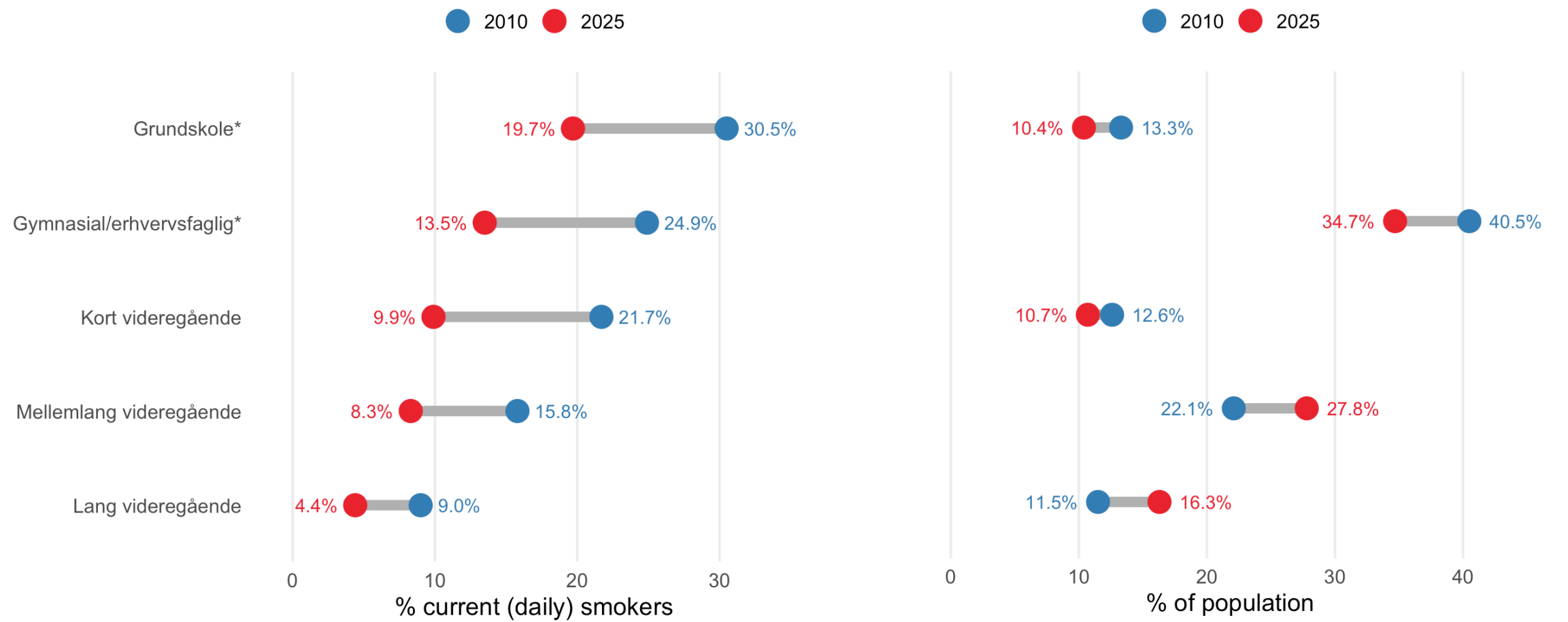
Which ‘components’?

Bridge between descriptive and causal

What if?

Techniques often involve ‘counterfactual’ scenarios

Danish smoking has declined from 21% (2010) to 11% (2025)



Source: Den Nationale Sundhedsprofil 2010 (Tabel 4.1.1) and 2025 (Tabel 3.1.3), Statens Institut for Folkesundhed / Sundhedsstyrelsen.

Covered Today

- 1. Kitagawa's Decomposition**
- 2. Oaxaca–Blinder Decomposition**
- 3. Concentration Index Decomposition**
- 4. Interventional Decomposition**

1. Kitagawa's Decomposition

COMPONENTS OF A DIFFERENCE BETWEEN TWO RATES*

EVELYN M. KITAGAWA

University of Chicago and Scripps Foundation

WHEN comparing the incidence of some phenomenon in two or more groups, social researchers place much emphasis on the need for holding constant those related factors that would tend to distort the comparison. For example, before comparing the death rates for the residents of two areas, demographers frequently control the factors of differences between the areas in age, sex and race composition. A tech-

If standardization alters a difference between two rates, it should be possible ... to break it up into **components** attributable to the various factors for which the data were standardized.

‘Unpacking’ the difference between crude rates of two groups

Potential consequences of ignoring composition

Impact of upward social mobility on population mortality: analysis with routine data

Richard F Heller, Patrick McElduff, Richard Edwards

may give a distorted view of the overall level of health inequality due to social class differentials as it does not take into account the changing distribution of social class within the whole population.

changes in the distribution of social class across the population are an important contributor to recent reductions in mortality

Group rate: $R_i = y_i / n_i$

Group share: $P_i = n_i / \sum_i n_i$

Crude rate: $\frac{\sum_{i=1}^I y_i}{\sum_{i=1}^I n_i} = \sum_{i=1}^I P_i R_i$

Standardized rate:

$$\frac{\sum_{i=1}^I y_i}{\sum_{i=1}^I s_i} = \sum_{i=1}^I S_i R_i$$

where S_i is the share of group i in a *standard* population ($\sum_i S_i = 1$)

Data for crude rates

Stratum	Pop	Count	Rate
1	n_i	y_i	r_i
$\vdots$	$\vdots$	$\vdots$	$\vdots$
I	N	Y	R

	Group A		Group B	
Age	Share	Rate	Share	Rate
Young	30%	10	50%	20
Old	70%	40	50%	30
Crude		31		25
Standardized to A		31		27

Standardization question:

What would the $A - B$ difference be if B had A 's age distribution?

Kitagawa's key insight



...if the difference between two standardized rates is subtracted from the corresponding difference in crude rates...the result is a weighted average of differences between the composition of the two groups.

But what are the weights?

$$\text{Crude difference} = \sum P_{Ai} R_{Ai} - \sum P_{Bi} R_{Bi}$$

→ Reflects differences in both rates *and* composition

$$\text{A-Standardized difference} = \sum P_{Ai} R_{Ai} - \sum P_{Ai} R_{Bi}$$

→ Reflects *only* differences in rates (P_{Ai} is constant)

$$\text{Crude - Standardized} = \sum R_{Bi} (P_{Ai} - P_{Bi})$$

→ Differences in composition, *weighted* by B s rates!

Applying the decomposition

$$\underbrace{\text{Rate}_A - \text{Rate}_B}_{\text{crude difference}} = \overbrace{\sum_i \left(\frac{R_{Ai} + R_{Bi}}{2} \right) (P_{Ai} - P_{Bi})}^{\text{composition effect}} + \underbrace{\sum_i \left(\frac{P_{Ai} + P_{Bi}}{2} \right) (R_{Ai} - R_{Bi})}_{\text{rate effect}}$$

R_{Ai} is the rate for group A in stratum i .

P_{Ai} is the i -stratum-specific weight in group A.

Age	Group A		Group B	
	Share	Rate	Share	Rate
Young	30%	10	50%	20
Old	70%	40	50%	30

Composition effect			
Group	$\frac{R_{Ai} + R_{Bi}}{2}$	$P_{Ai} - P_{Bi}$	Contribution
Young	15	-0.20	-3.0
Old	35	+0.20	+7.0
Sum			+4.0

Rate effect			
Group	$\frac{P_{Ai} + P_{Bi}}{2}$	$R_{Ai} - R_{Bi}$	Contribution
Young	0.40	-10	-4.0
Old	0.60	+10	+6.0
Sum			+2.0

Total = **+6.0** = crude gap (31 – 25) → **67% due to composition**

Rate effect (+2) = standardized gap (avg reference) (28 – 26)

Kitagawa: does the reference group matter?

Reference	Composition	Rate	Total
A's rates	6.0	0.0	6.0
B's rates	2.0	4.0	6.0
Average (Kitagawa)	4.0	2.0	6.0

Crude gap = $31 - 25 = 6$ in every row; only the split changes

The one-sided rows differ by the interaction, $\sum_i (P_{Ai} - P_{Bi})(R_{Ai} - R_{Bi}) = 4.0$, which the average splits evenly

What accounts for differences in neonatal mortality rates across European countries?

GA-specific rates or distribution?

Implications for interventions?

Sartorius et al. (2024)

Original Investigation | Pediatrics

Neonatal Mortality Disparities by Gestational Age in European Countries

Victor Sartorius, MD; Marianne Philibert, MSc; Kari Klungsoyr, PhD; Jeannette Klimont, MSc; Katarzyna Szamotulska, PhD; Zeljka Drausnik, MPH; Petr Velebil, PhD; Laust Mortensen, PhD; Mika Gissler, PhD; Jeanne Fresson, MD; Jan Nijhuis, PhD; Wei-Hong Zhang, PhD; Karin Källén, PhD; Tonia A. Rihs, PhD; Vlad Tiza, PhD; Ruth Matthews, PhD; Lucy Smith, PhD; Jennifer Zeitlin, DSc; for the Euro-Peristat Network

Abstract

IMPORTANCE There are wide disparities in neonatal mortality rates (NMRs, deaths <28 days of life after live birth per 1000 live births) between countries in Europe, indicating potential for improvement. Comparing country-specific patterns of births and deaths with countries with low mortality rates can facilitate the development of effective intervention strategies.

OBJECTIVE To investigate how these disparities are associated with the distribution of gestational age (GA) and GA-specific mortality rates.

DESIGN, SETTING, AND PARTICIPANTS This was a cross-sectional study of all live births in 14 participating European countries using routine data compiled by the Euro-Peristat Network. Live births with a GA of 22 weeks or higher from 2015 to 2020 were included. Data were analyzed from May to October 2023.

EXPOSURES GA at birth.

MAIN OUTCOMES AND MEASURES The study investigated excess neonatal mortality, defined as a rate difference relative to the pooled rate in the 3 countries with the lowest NMRs (Norway, Sweden, and Finland; hereafter termed the top 3). The Kitagawa method was used to divide this excess into the proportion explained by the GA distribution of births and by GA-specific mortality rates. A sensitivity analysis was conducted among births 24 weeks' GA or greater.

RESULTS There were 35 094 neonatal deaths among 15 123 428 live births for an overall NMR of 2.32 per 1000. The pooled NMR in the top 3 was 1.44 per 1000 (1937 of 1 342 528). Excess neonatal mortality compared with the top 3 ranged from 0.17 per 1000 in the Czech Republic to 1.82 per 1000 in Romania. Excess deaths were predominantly concentrated among births less than 28 weeks' GA (57.6% overall). Full-term births represented 22.7% of the excess deaths in Belgium, 17.8% in France, 40.6% in Romania and 17.3% in the United Kingdom. Heterogeneous patterns were observed when partitioning excess mortality into the proportion associated with the GA distribution vs GA-specific mortality. For example, these proportions were 9.2% and 90.8% in France, 58.4% and 41.6% in the United Kingdom, and 92.9% and 7.1% in Austria, respectively. These associations remained stable after removing births under 24 weeks' GA in most, but not all, countries.

CONCLUSIONS AND RELEVANCE This cohort study of 14 European countries found wide NMR disparities with varying patterns by GA. This knowledge is important for developing effective strategies to reduce neonatal mortality.

JAMA Network Open. 2024;7(8):e2424226. doi:10.1001/jamanetworkopen.2024.24226

Key Points

Question Are disparities in neonatal mortality between European countries associated with differences in the gestational age distribution of births or with higher mortality at specific gestational ages?

Findings In this cross-sectional study of over 15 million live births from 14 European countries, heterogeneous patterns were observed across countries and gestational age groups in the association of these 2 components with excess mortality in comparison with the 3 countries with the lowest neonatal mortality rates (Sweden, Norway, and Finland).

Meaning Stratifying neonatal mortality rates by gestational age can provide country-specific insights to inform preventive strategies to reduce neonatal mortality.

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

Hiding in plain sight?

Both DK and AT similar and ~0.3 deaths per 1000 higher than “Top 3”

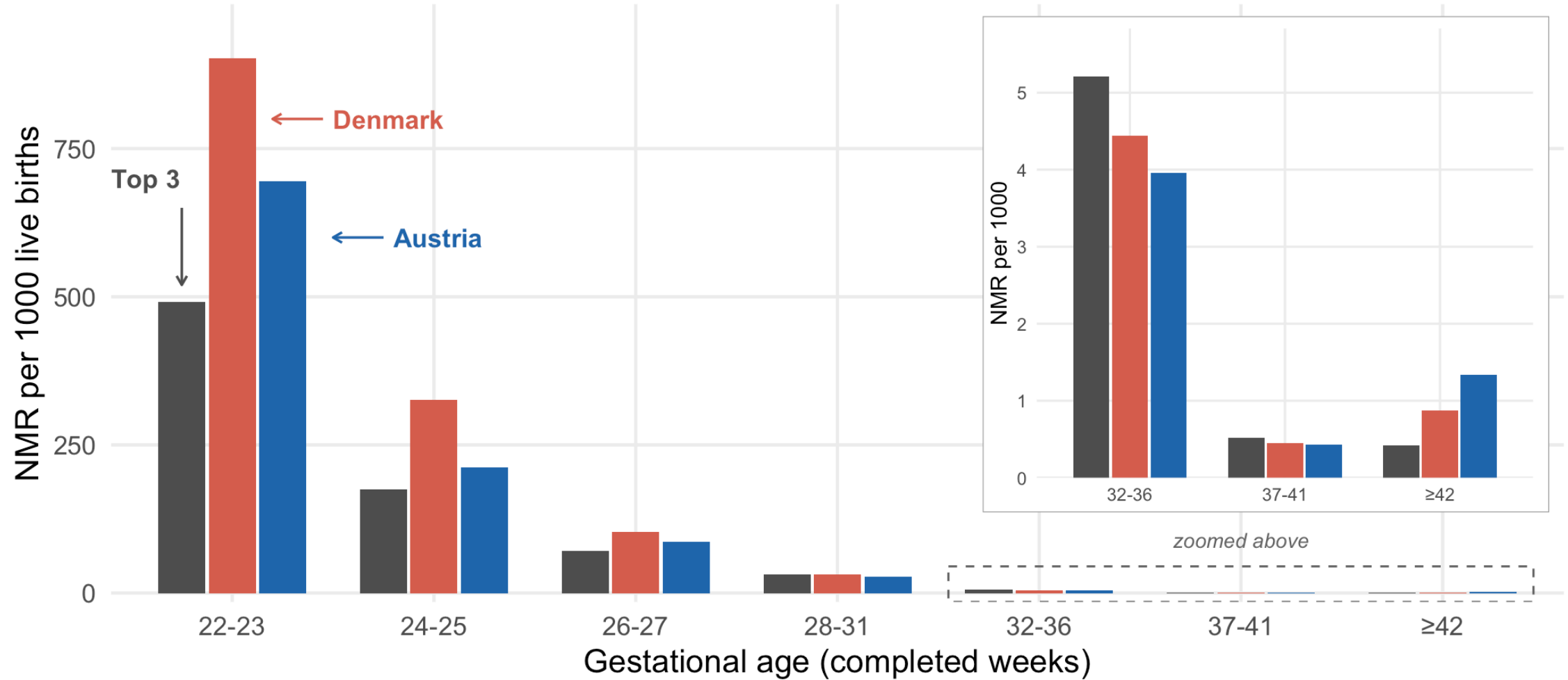
But *completely different* contributions to the difference

Table 3. Neonatal Mortality Rate (NMR) Differences for Each Country With Sweden, Norway, and Finland and the Kitagawa Decomposition Into the Differences Attributed to the Gestational Age (GA) Distribution and to GA-Specific NMRs Among Live Births at 22 Weeks or More and 24 Weeks or More of GA^a

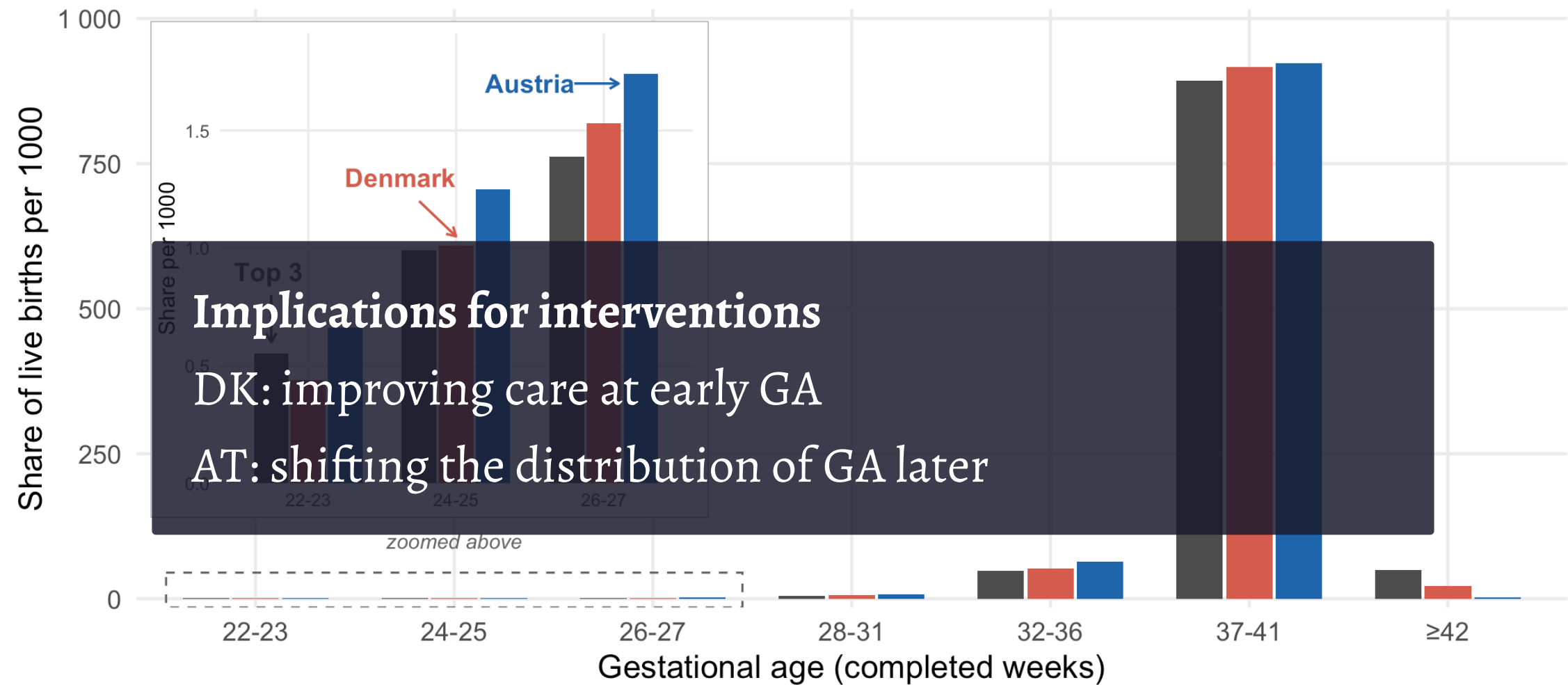
Country	NMR per 1000 live births	Difference from Finland, Norway, and Sweden	% of the Difference attributed to	
			GA distribution	GA-specific NMR
Live births ≥22 weeks of GA				
Finland, Norway, and Sweden	1.44	NA	NA	NA
Czech Republic	1.61	0.17	27.7	72.3
Austria	1.73	0.28	92.9	7.1
Denmark	1.74	0.30	0.0	100.0

Sartorius et al. (2024)

Denmark has much worse rates at early GA



Austria more births at earlier GA



Sartorius et al. (2024)

Summary: Kitagawa's decomposition

When to use

Comparing rates across groups or over time when their composition differs

Extensions

More than two groups or factors; three-part decomposition

Caveats

Not causal; results depend on the strata used and on the reference weights

Covered Today

1. **Kitagawa's Decomposition**
2. **Oaxaca–Blinder Decomposition**
3. **Concentration Index Decomposition**
4. **Interventional Decomposition**

2. Oaxaca–Blinder Decomposition

Oaxaca-Blinder: Basic Idea

What explains average differences in outcomes? In a regression context:

1. Means

Differences in the prevalence of determinants of outcome

2. Coefficients

Differences in the coefficient of a given determinant on the outcome (i.e., effect measure modification)

Inequalities in the use of health services between immigrants and the native population in Spain: what is driving the differences?

Dolores Jiménez-Rubio · Cristina Hernández-Quevedo

Our findings show that **unexplained factors associated to immigrant status** determine to a great extent disparities in the probability of using hospital, specialist and emergency services of immigrants relative to Spaniards, while individual characteristics, in particular self-reported health and chronic conditions, are much more important in explaining the differences in the probability of using general practitioner services between immigrants and Spaniards

Origins

RESEARCH ARTICLE

Oaxaca-Blinder meets Kitagawa: What is the link?

Ronald L. Oaxaca^{1,2,3}, Eva Sierminska^{1,2,3,4,5,6}

¹ Department of Economics, University of Arizona, Tucson, Arizona, United States of America, ² IZA, Bonn, Germany, ³ GLO, Essen, Germany, ⁴ Luxembourg Institute of Socio-Economic Research (LISER), Esch/Alzette, Luxembourg, ⁵ INE PAN, Warsaw, Poland, ⁶ DIW, Berlin, Germany

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● These authors contributed equally to this work.

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Studies by R. Oaxaca (1973) and Blinder (1973) applied regression-based decomposition methods to analyze the wage gap between men and women and between whites and blacks in the USA.

Focused on how much of wage gap was ‘explained’ by differences in observable characteristics

Recent attempts to reconcile O-B with Kitagawa (equivalent for binary outcome and categorical covariates)

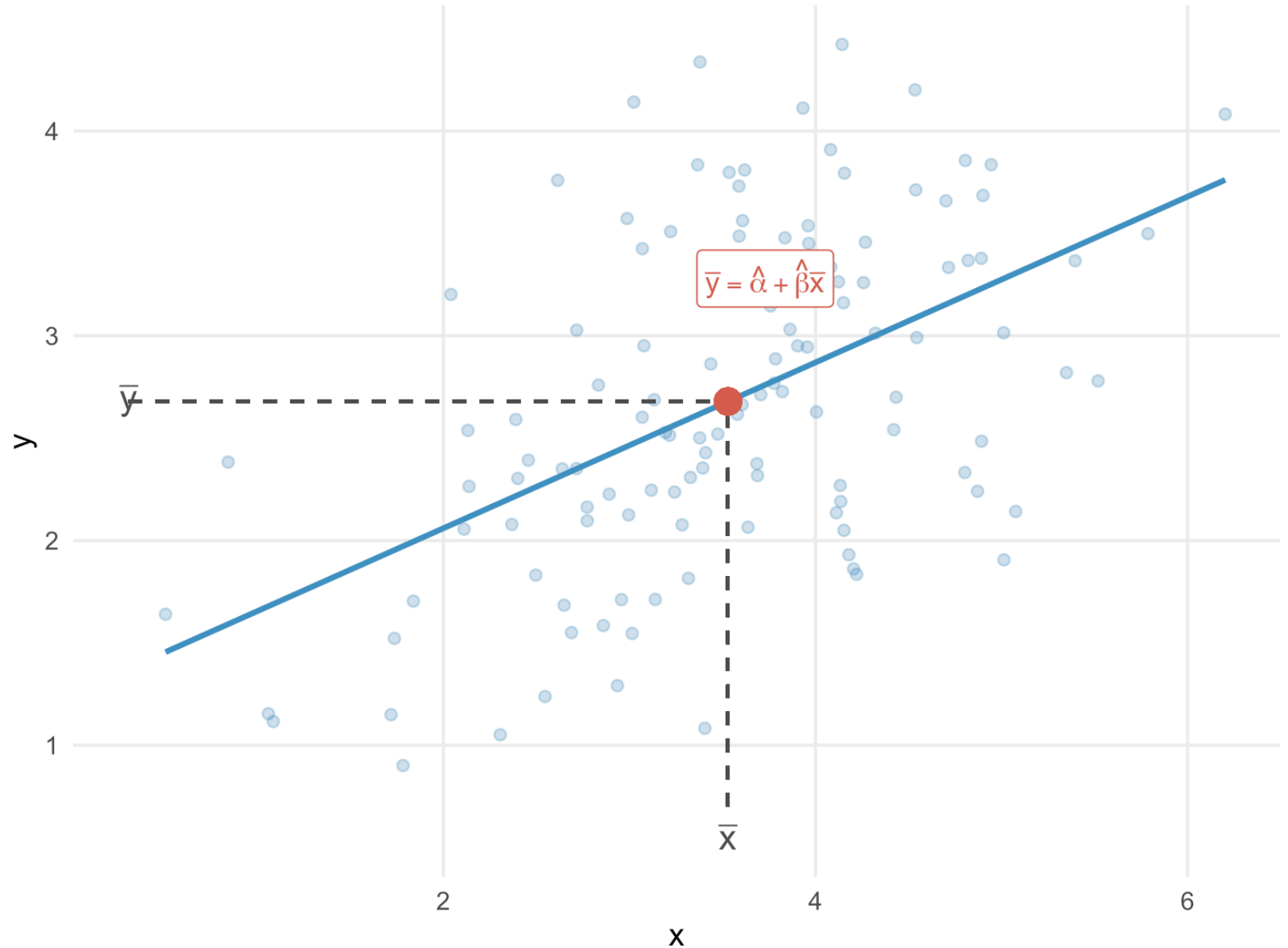
Brief note on interpretation

Decomposition methods are based on regression analyses, and thus all of the usual caveats about good specification apply.

If regressions are purely descriptive, they reveal the associations that characterize the health inequality. Then inequality is explained in a statistical sense but implications for policies to reduce inequality are limited.

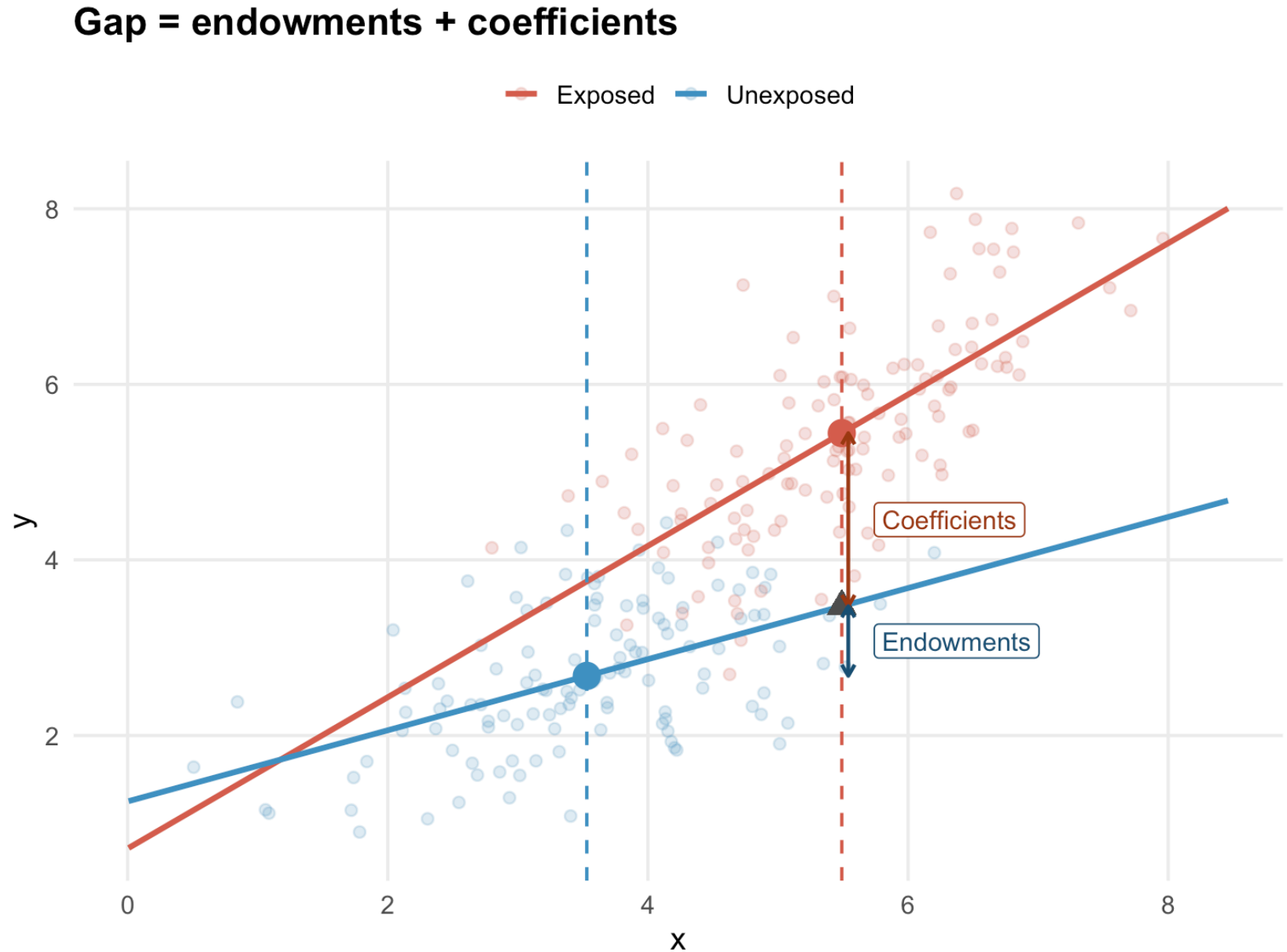
Why does this
work?

OLS regression
line always passes
through $(\bar{x}, \bar{y})$
because residuals
sum to zero.



Why does this work?

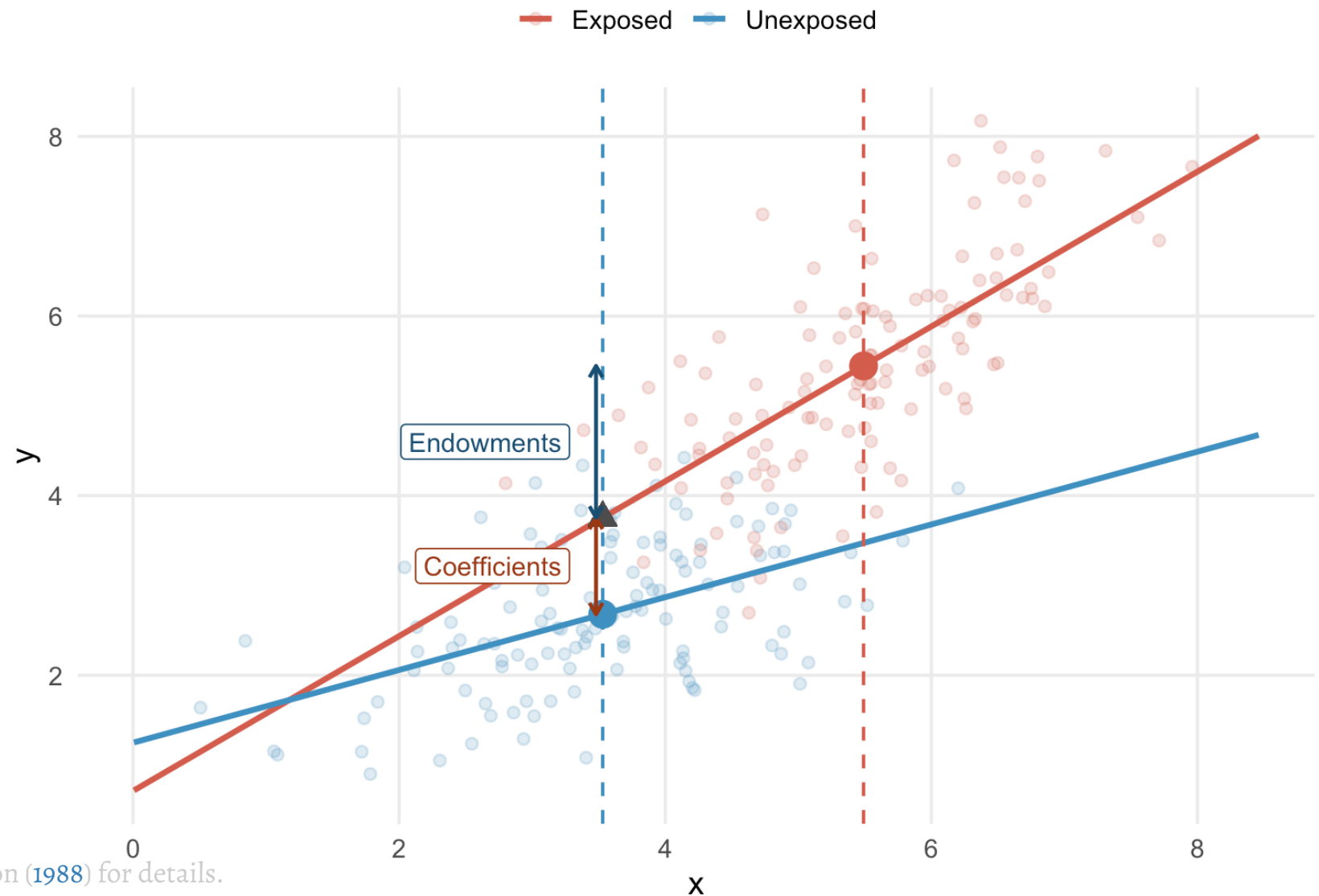
The gap decomposes into **endowments** (different $\bar{x}$, same coefficients) and **coefficients** (same $\bar{x}$, different slopes/intercepts).



Equally valid to
use the **exposed**
group's slope

Same total gap,
but different split

Same gap, alternative split (exposed slope as reference)



See R. Oaxaca (1973), Blinder (1973) and Cotton (1988) for details.

Example: Educational Differences in BMI in Italy

What is the average difference in body mass index (BMI) between those with low vs. high education?

How much is due to *differing determinants* of BMI (age, gender, smoking, drinking, income)?

Any residual difference is due to education differences in the *associations* between those risk factors and BMI — i.e., the coefficients differ.

Example data

European Social Survey Round 11, Italy (n = 1743)

Body mass index as outcome (kg/m²): `weight / height2`

Overall difference by education: High ed (ISCED 5–7) vs Low ed (ISCED 1–4)

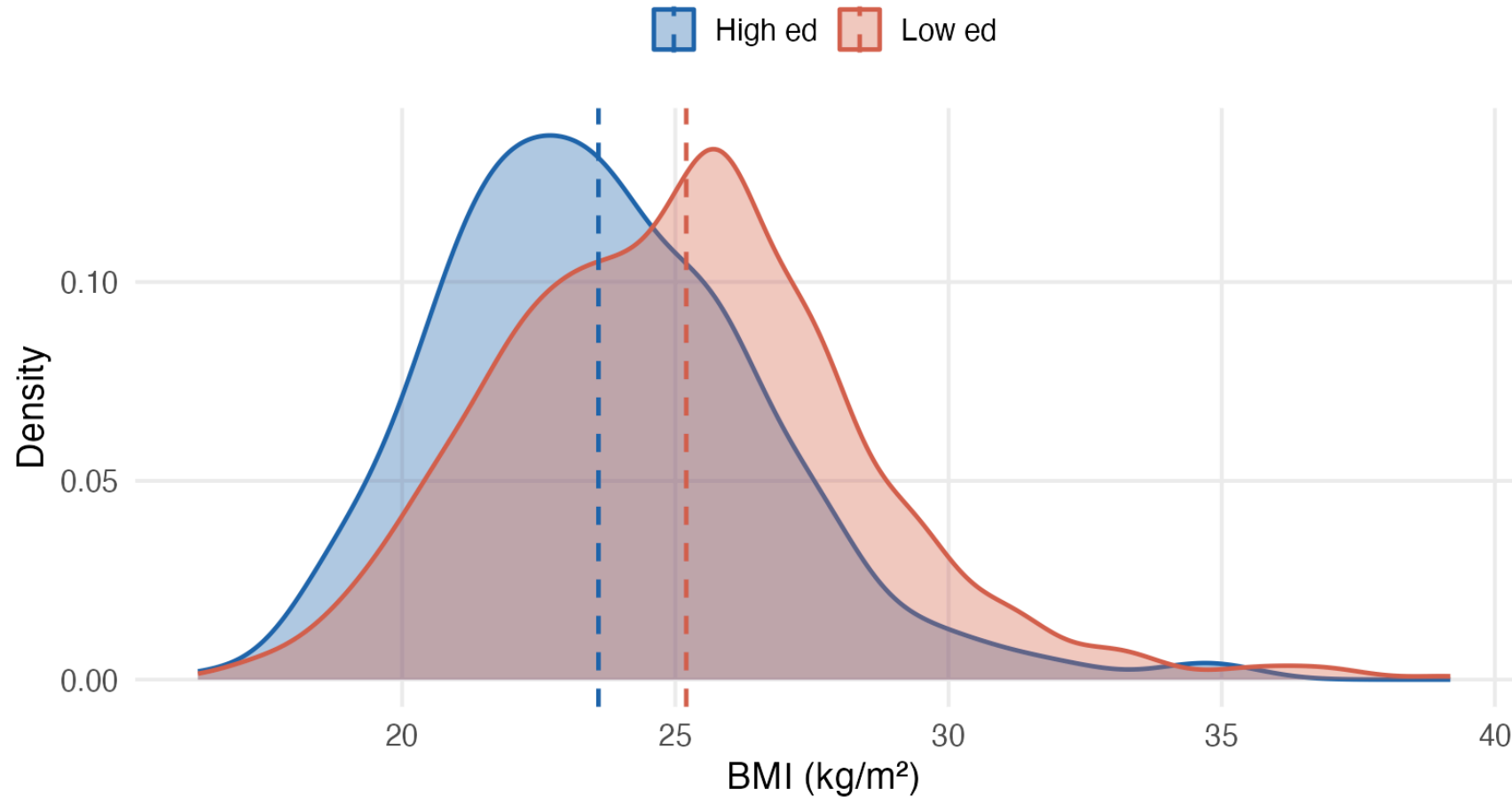
Potential determinants (the *X*s):

- age (years)
- gender (female = 1)
- smoking (1 = current smoker)
- binge drinking (1 = monthly or more)
- married (1 = currently married)
- income decile (1–10)

High ed BMI: 23.6, Low ed BMI: 25.2, Gap = 1.6

Distribution of BMI (kg/m²) by low_ed

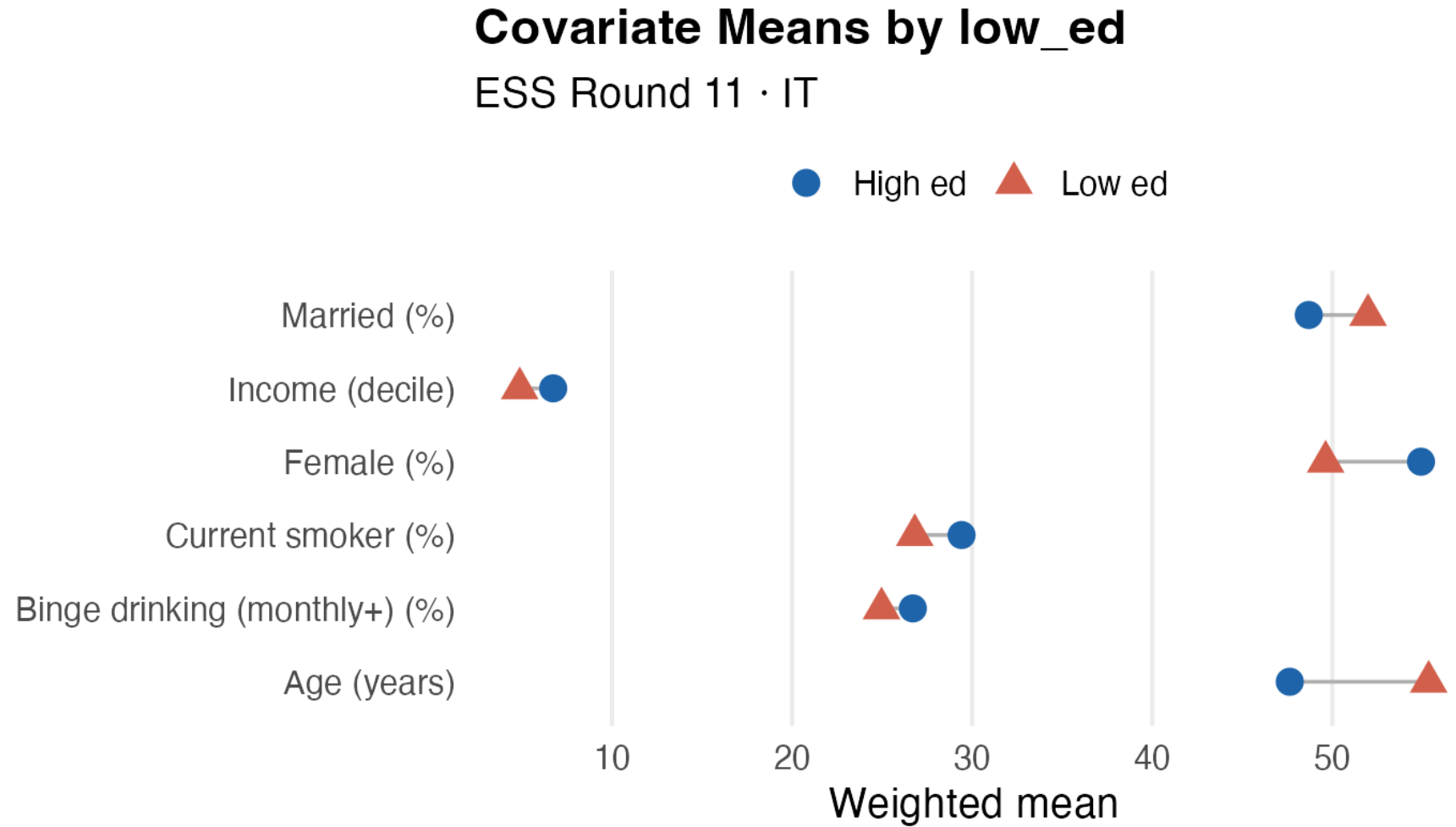
ESS Round 11 · IT · n = 1743



Source: Author's calculations

Differences in determinants

Differences in age, smoking, etc. could explain part of the BMI gap



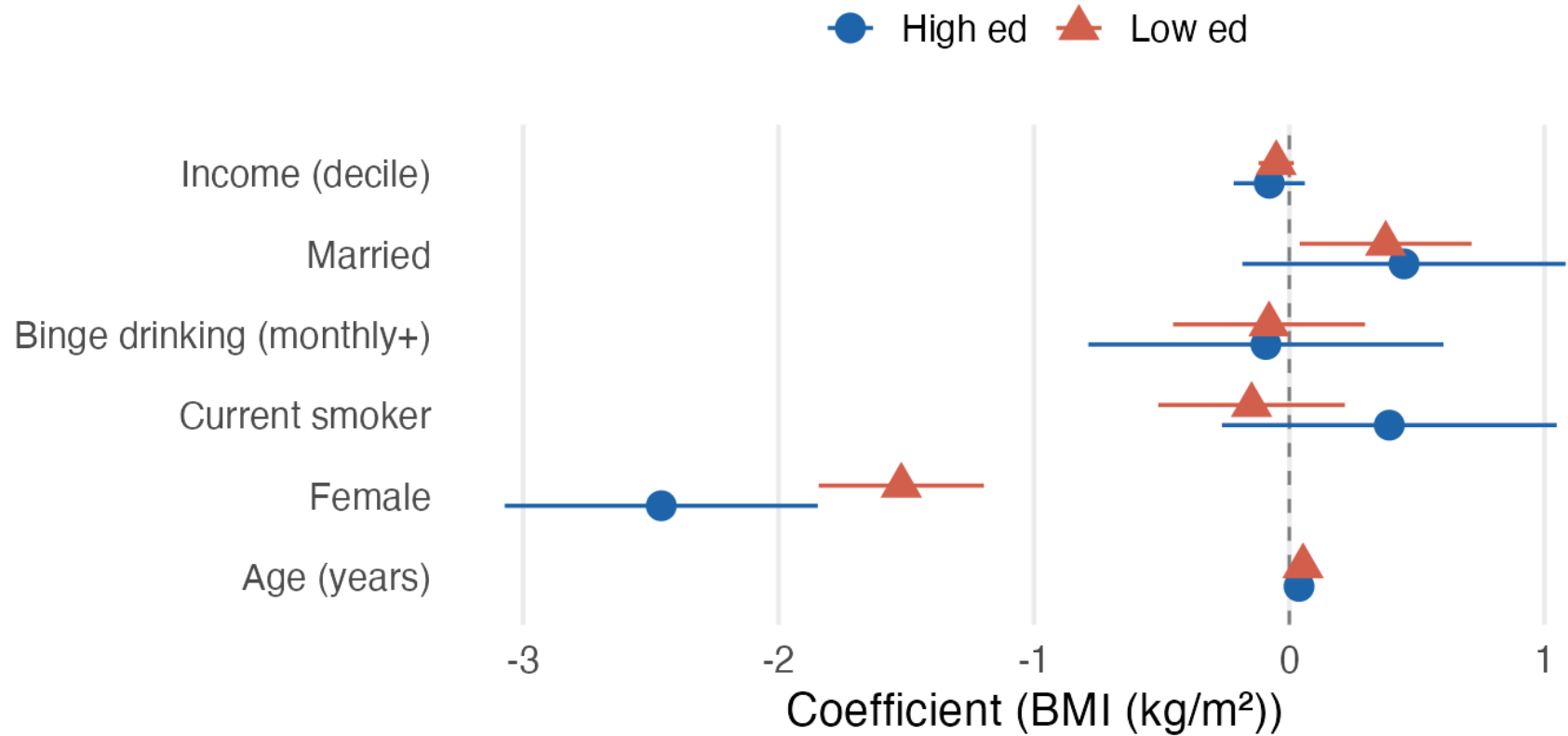
Source: Author's calculations

Differences in coefficients

- Each covariate may have a different association with BMI by education
- Differences in *returns* form the unexplained component

Regression Coefficients by low_ed

Outcome: BMI (kg/m²) · ESS Round 11 · IT



Decomposition results: variable contributions

Overall gap = 1.61 kg/m². Endowments: 38%; Coefficients: 62%

	Endowments		Coefficients	
	Est	SE	Est	SE
Total	0.608	0.108	1.030	0.259
Age (years)	0.415	0.074	0.865	0.691
Female	0.080	0.051	0.466	0.181
Current smoker	0.004	0.008	-0.145	0.115
Binge drinking (monthly+)	0.001	0.007	0.003	0.104
Married	0.012	0.016	-0.037	0.208
Income (decile)	0.095	0.071	0.133	0.425
Intercept	—	—	-0.256	0.837

Note: Low ed betas used as reference

Summary: Endowments = 38%, Coefficients = 62%

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If the low-educated had the same covariate means as the high-educated (keeping the low-educated group's own coefficients), their BMI would be 0.608 kg/m^2 lower — closing 38% of the gap.

Most of this is due to older age among the low-educated, which predicts higher BMI.

Summary: Endowments = 38%, Coefficients = 62%

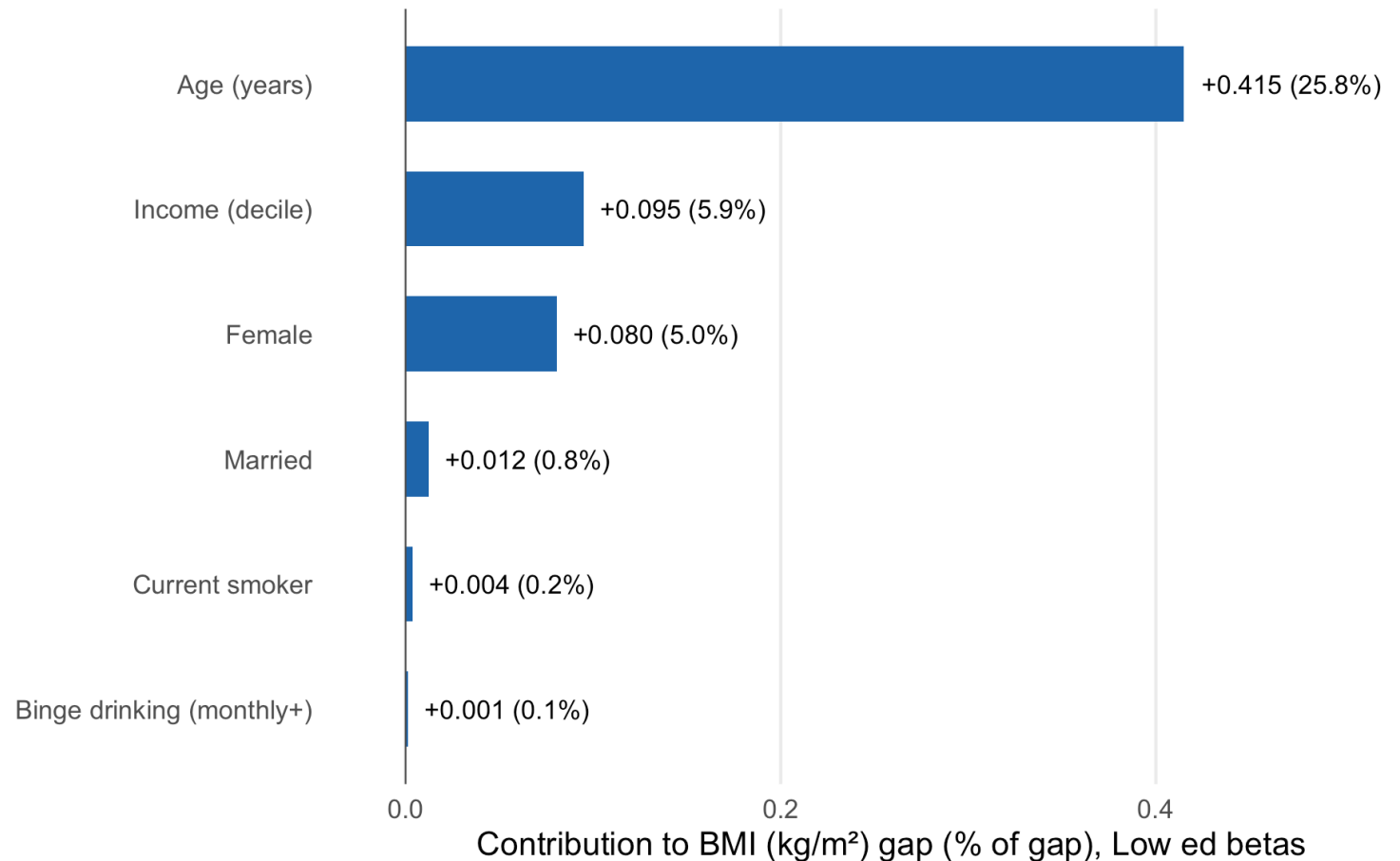
If the high-educated group's covariates had the same relationship with BMI as they do in the low-educated group, their BMI would be 1.030 higher, accounting for 62% of the gap.

Smoking is *negative*, since smoking predicts higher BMI among the high-educated and is more common among high educated.

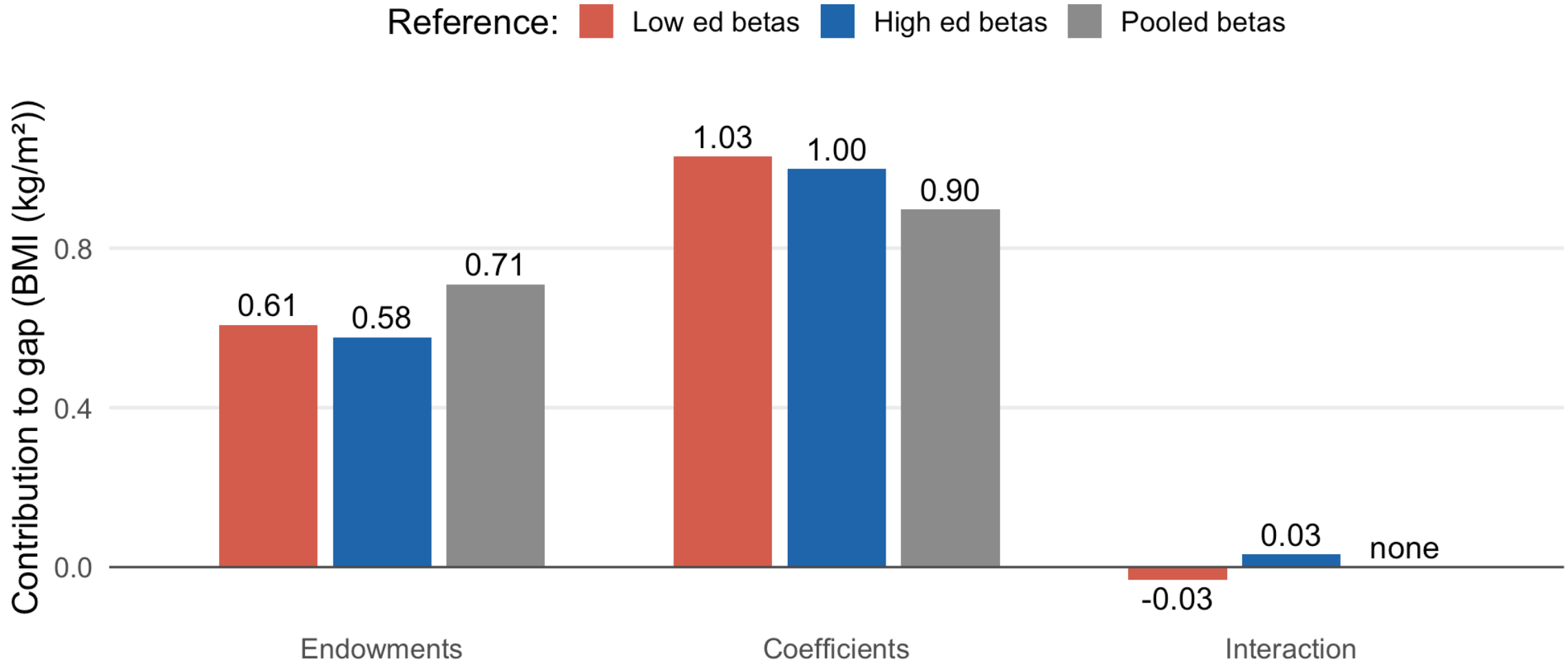
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Which covariates drive the endowments?

- Variables where groups differ *and* that predict BMI contribute most
- Income and age are the biggest drivers



Decomposition results: does the reference group matter?



Endowments account for 36–44% of it; interaction is negligible

Summary: Oaxaca–Blinder decomposition

When to use

Explaining average gaps with potential determinants

Extensions

Nonlinear outcomes (binary, counts); pooled reference coefficients; variable-by-variable contributions

Caveats

Not causal; split depends on the reference group and, for categorical covariates, the omitted category

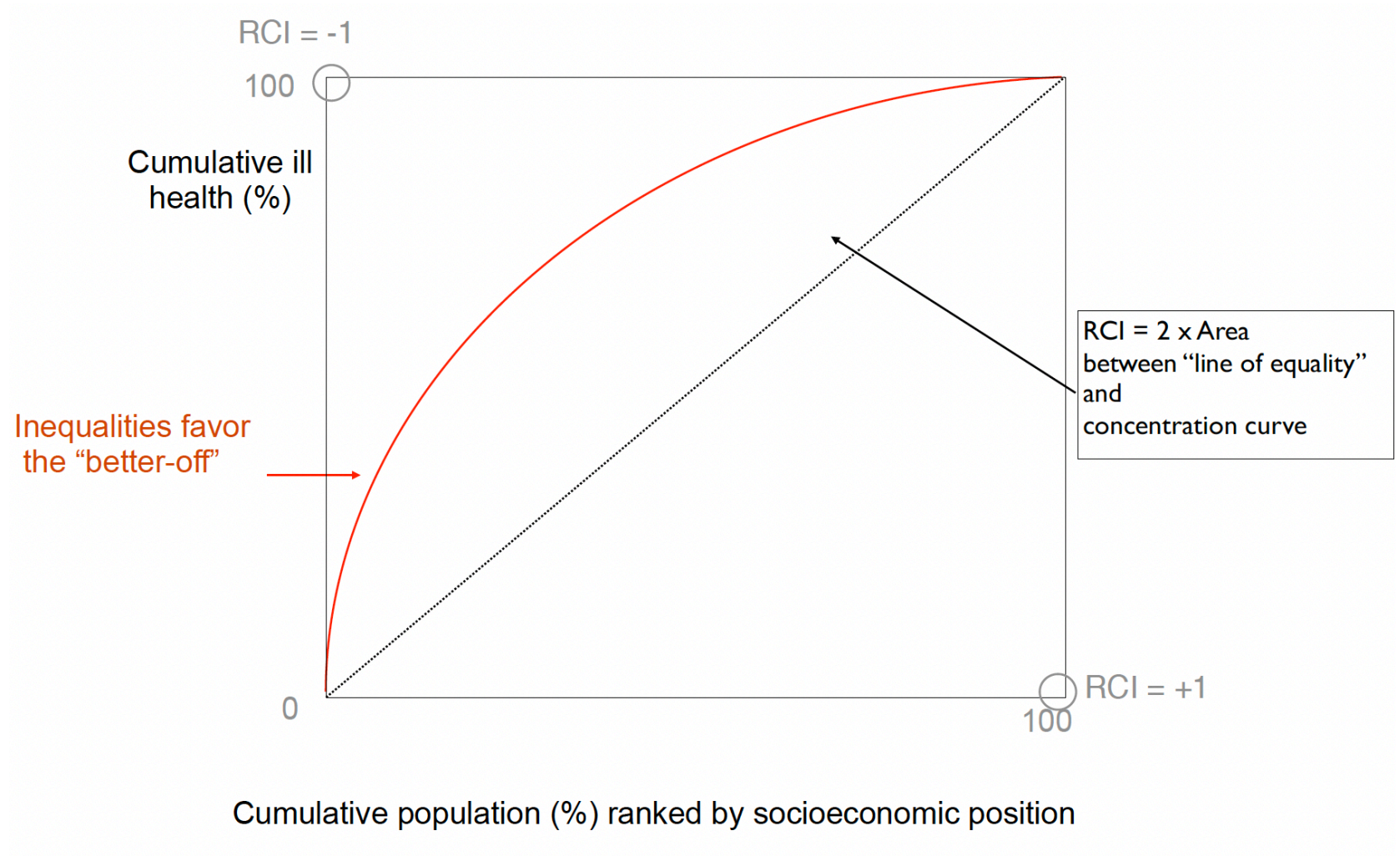
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3. Concentration Index Decomposition

Relative Concentration Curve

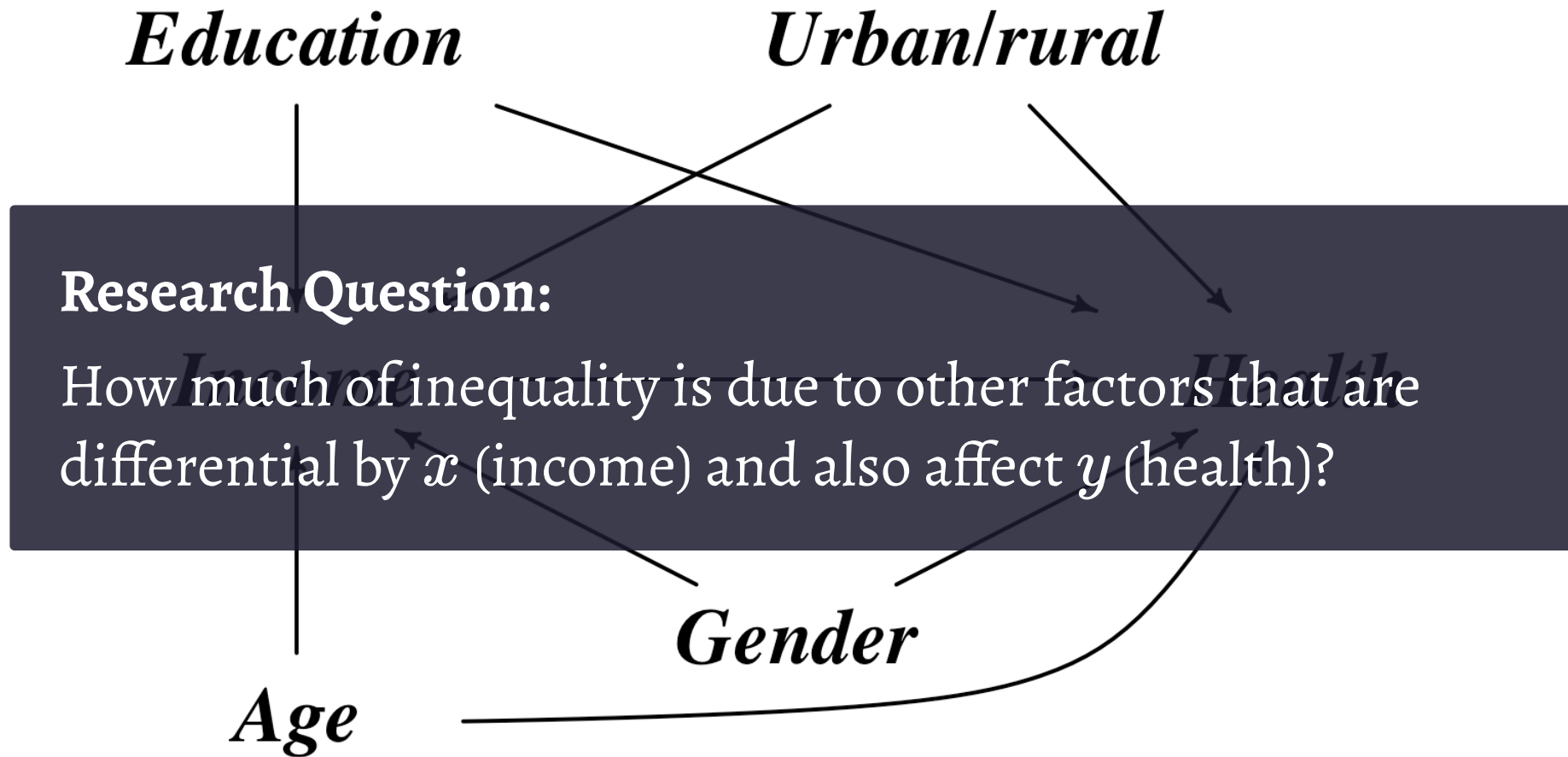
Explicitly accounts for changing composition



We want to understand this:

Income —————→ *Health*

By evaluating something like this:



Relative Concentration Index

$$RCI = \frac{2}{n\mu} \sum_{i=1}^n y_i R_i - 1$$

where

- μ is the mean of y_i (e.g., smoking);
- R_i is the fractional rank of the i th person in the socioeconomic (i.e., income) distribution.

Decomposition:

Develop a **model for predicting y** using several determinants, then plug back into the *RCI* equation

Estimates how much of the overall inequality in y is due to the association between income and other factors that predict health

RCI

Decomposition

RCI is a function of health (y_i) and socioeconomic rank (R_i), i.e.

$$RCI = \frac{2}{n\mu} \sum_{i=1}^n y_i R_i - 1$$

Then write a regression expressing health (y_i) as a function of several k_i determinants (e.g., age, gender, urban/rural status):

$$y_i = \alpha + \sum \beta_x x_{k_i} + \epsilon_i$$

RCI Decomposition

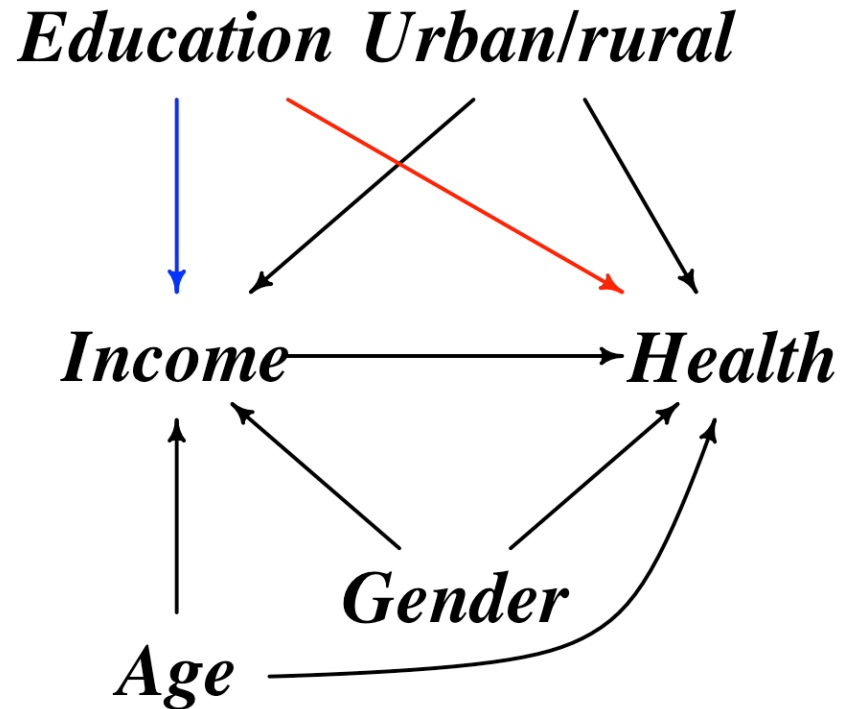
Now re-express RCI as:

$$RCI = \sum (\beta_k \bar{x}_k / \mu) RCI_k + gRCI_e / \mu$$

Where

- μ is the mean of y ,
- $\bar{x}_k$ is the mean of determinant x_k ,
- β_k is the regression coefficient for x_k , and
- RCI_k is the relative concentration index for x_k .

Two types of 'explained' components



Determinant impact depends on:

1. the strength of the relationship between each factor and income RCI_k
2. the strength of the relationship between each factor and health, and its prevalence in the population (elasticity) $\beta_k \bar{x}_k / \mu$

Procedure for decomposing the Concentration Index

1. Regress y ('health') on its determinants ($\beta_k x_k$):

$$y_i = \alpha + \sum \beta_x x_{k_i} + \epsilon_i$$

2. Calculate means of y (μ) and all x_k determinants
3. Calculate (CI) for the health variable *and* for each determinant (CI_k). This means using each determinant x_k as the “outcome” and estimate a CI for age, CI for education, etc.

Procedure for decomposing the Concentration Index

4. Absolute contribution: for each x multiply 'elasticity' by its concentration index (CI_k):

$$(\beta_k \bar{x}_k / \mu) RCI_k$$

5. Calculate the % contribution of each determinant:

$$[(\beta_k \bar{x}_k / \mu) RCI_k] / RCI$$

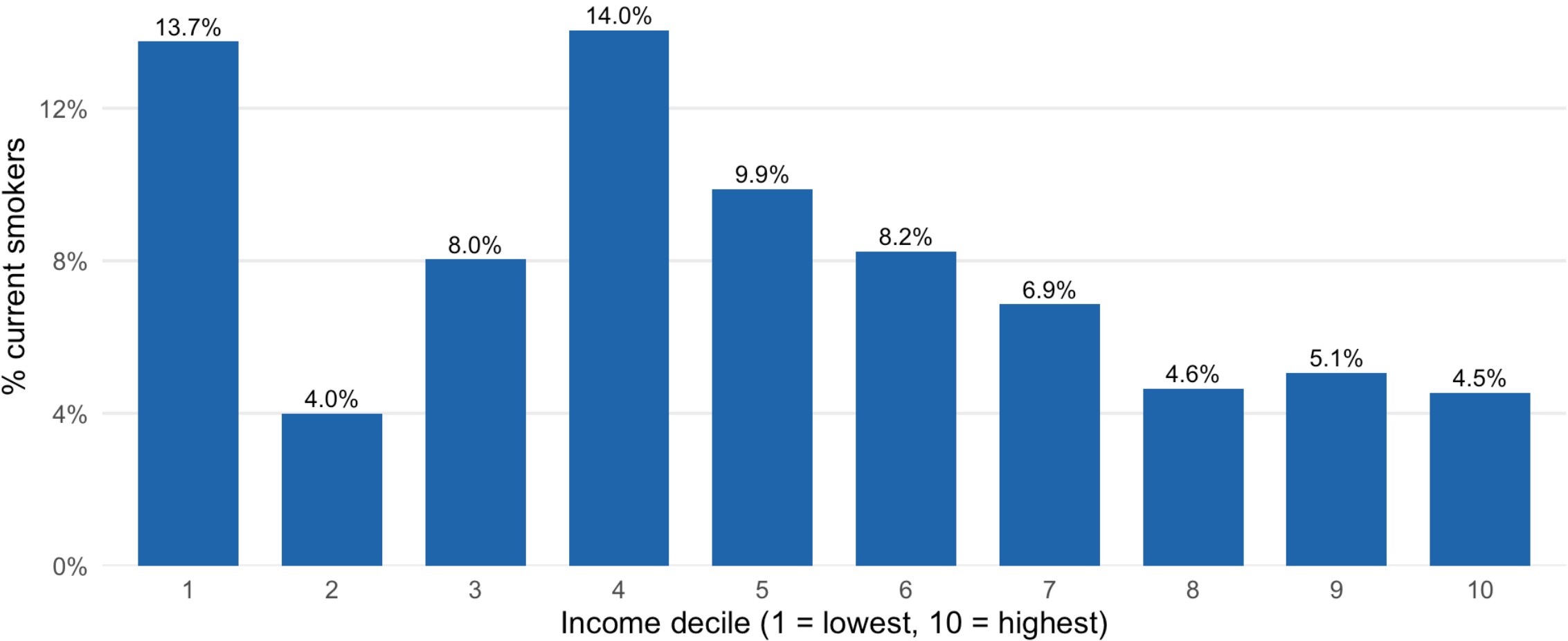
Example: Decomposing Socioeconomic Inequality in Current Smoking

Data: European Social Survey Round 11 for Sweden

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
Current smoker	2	0	0.1	0.2	0.0	0.0	1.0	
Income decile	10	0	6.6	2.6	1.0	7.0	10.0	
Age (years)	76	0	54.4	18.6	15.0	56.0	90.0	
Low education (ISCED 1–4)	2	0	0.4	0.5	0.0	0.0	1.0	
Female	2	0	0.5	0.5	0.0	0.0	1.0	
Binge drinking (monthly+)	2	0	0.5	0.5	0.0	0.0	1.0	
Obese (BMI ≥ 30)	2	0	0.1	0.4	0.0	0.0	1.0	
Married	2	0	0.5	0.5	0.0	0.0	1.0	
Survey weight	132	0	1.0	0.5	0.3	0.8	4.0	

Smoking Prevalence by Income Decile

ESS Round 11 · Sweden · n = 1162 · Overall smoking rate = 7.3%



Source: Author's calculations

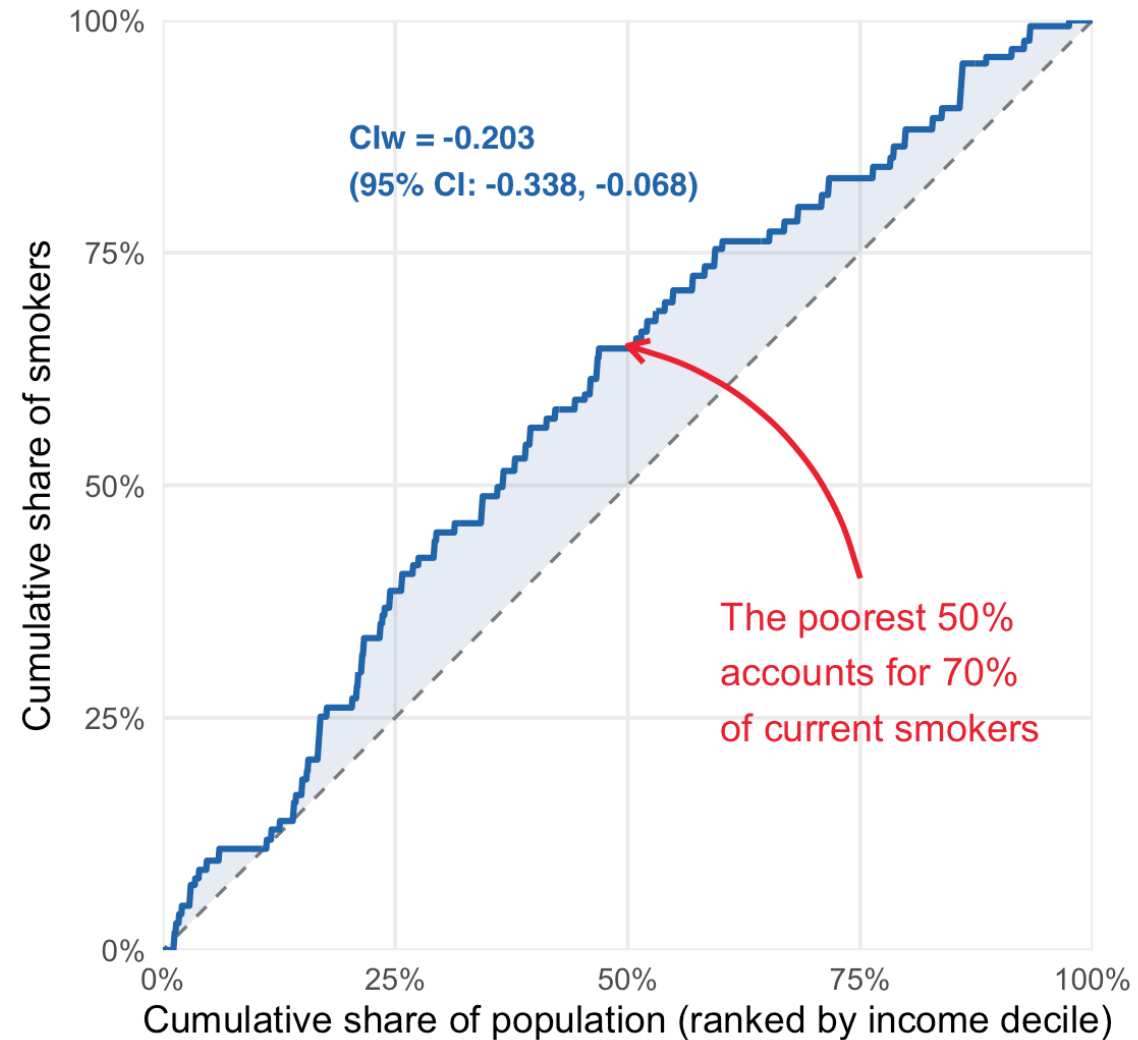
What are we explaining?

Overall CI = -0.20

Smoking *more concentrated* among the poor

Concentration Curve: Current Smoking by Income Decile

ESS Round 11 · SE · n = 1162



Step 1: Predictors of current smoking

Variable	β	OR	95% CI	p
Age (years)	-0.001	0.999	(0.99, 1.01)	0.906
Female	0.141	1.152	(0.73, 1.83)	0.547
Low education (ISCED 1–4)	0.359	1.432	(0.90, 2.27)	0.130
Married	-0.989	0.372	(0.21, 0.65)	<0.001
Binge drinking (monthly+)	0.38	1.463	(0.92, 2.32)	0.107
Obese (BMI $\geq$ 30)	0.495	1.64	(0.90, 2.99)	0.106

Step 2: Estimating elasticity for education

$$RCI = \sum \underbrace{(\beta_k \bar{x}_k / \mu)}_{\text{elasticity}} RCI_k + gRCI_e / \mu$$

- $\beta_{\text{low education}} = 0.36$ (OR = 1.4) Elasticity for low education is: $(0.024 * 0.452 / .073) = 0.148$
 - Marginal effect: 0.024
 - Mean low education: 0.452
 - Mean smoking rate: 7.3%
- Interpretation: a 1% increase in low education increases smoking by 14.8% (not percentage points!).

This is Step (2): Calculate the mean of y (μ) and of each of the x_k determinants

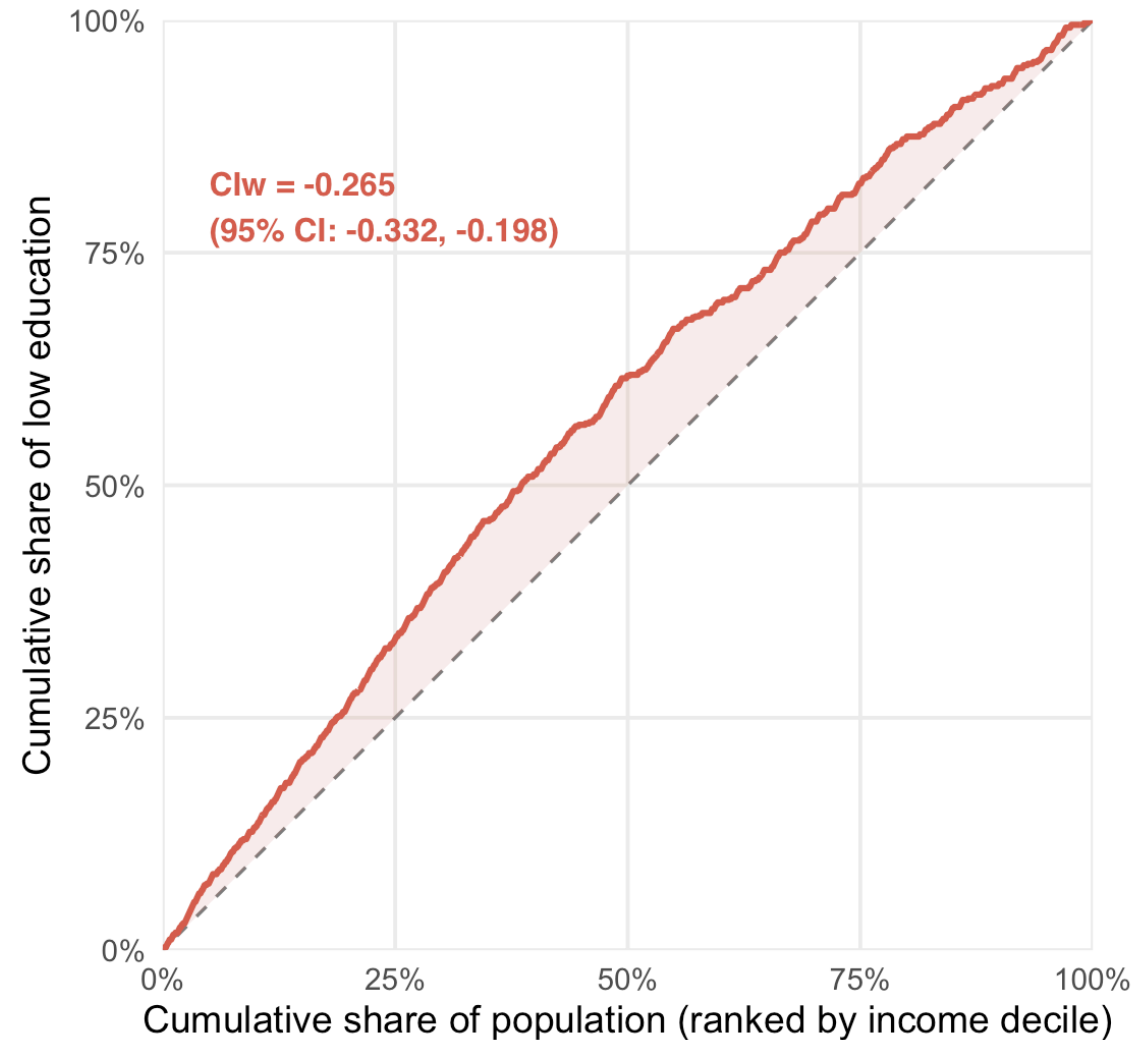
Step 3: RCI_k

Note the y-axis is cumulative share of low education

The poorest 50% account for roughly 60% of the share of low educated.

Concentration Curve: Low education by Income Decile

ESS Round 11 · SE · n = 1162



This is Step (3): Calculate the CI for each of the potential determinants x_k

Step 4: Calculate contributions

Estimation for a
specific factor:
Low education

$$RCI = \sum \underbrace{(\beta_k \bar{x}_k / \mu)}_{\text{elasticity}} RCI_k + gRCI_e / \mu$$

The elasticity of smoking (prior slide) = 0.148

Now we have RCI_k for low education = -0.265

The contribution of low education:

$$\text{Elasticity} \times RCI_{ed} = 0.148 * -0.265 = -.039$$

Thus low education accounts for $-.039 / -.203 = 19\%$ of the overall RCI

$$RCI = -0.203$$

(-) Binge drinking:
 ↑ smoking but
 more common at
 higher incomes

(+) Married: ↓
 smoking but more
 common at higher
 incomes

Variable	Elasticity	CI_k	Contribution	
			Absolute	%
Age	-0.030	0.001	-0.000	0.0
Binge drinker	0.152	0.103	0.016	-7.7
Female	0.060	-0.031	-0.002	0.9
Obese	0.068	-0.013	-0.001	0.4
Low education	0.144	-0.265	-0.038	18.8
Married	-0.353	0.284	-0.100	49.4
Residual	—	—	-0.077	38.1

Logistic regression; marginal effects used as elasticity weights.

Uncertainty in the RCI Decomposition

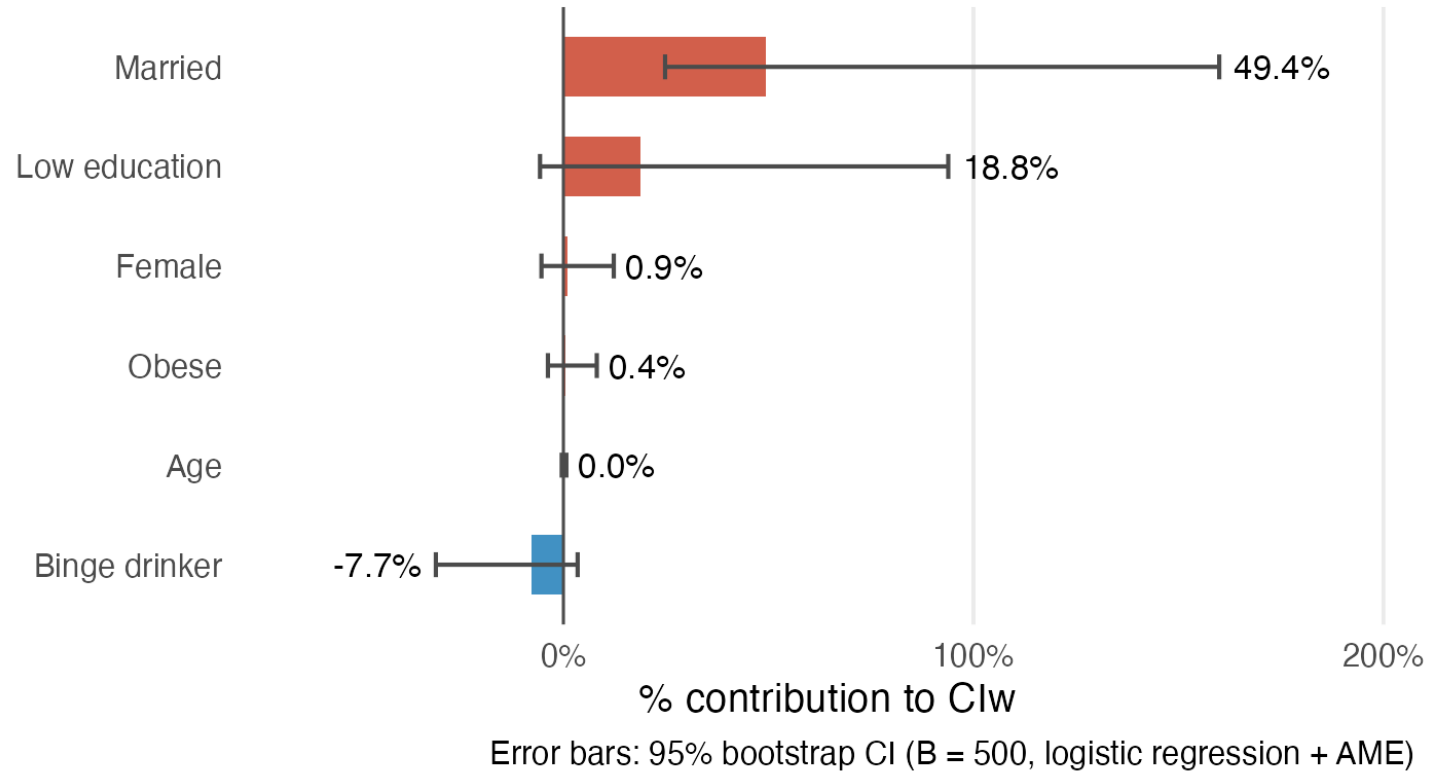
Each contribution has sampling variability:

$$\underbrace{\frac{\hat{\beta}_k \bar{x}_k}{\hat{\mu}}}_{\text{elasticity}} \times \underbrace{\widehat{RCI}_k}_{\text{inequality in } x_k}$$

Can **bootstrap** the decomp B times to get empirical CIs.

Decomposition of Smoking Inequality by Income

CIw = -0.203 | ESS Round 11 · SE · n = 1162



See O'Donnell et al. (2008); Hosseinpoor, Doorslaer, and Speybroeck (2006)

Summary: Concentration index decomposition

When to use

Summarizing with RCI and asking which determinants drive it

Extensions

Nonlinear models; Oaxaca-type decomposition of differences and temporal changes

Caveats

The explained share depends on which determinants enter; associations are predictive, not causal

Wagstaff, Doorslaer, and Watanabe (2003); Kakwani, Wagstaff, and Doorslaer (1997); O'Donnell et al. (2008); Erreygers and Kessels (2016)

Covered Today

- 1. Kitagawa's Decomposition**
- 2. Oaxaca–Blinder Decomposition**
- 3. Concentration Index Decomposition**
- 4. Interventional Decomposition**

4. Interventional Decomposition

Origins: Why ‘interventional’?

ORIGINAL ARTICLE

Decomposition Analysis to Identify Intervention Targets for Reducing Disparities

John W. Jackson,^{a,b,c} and Tyler J. VanderWeele^d

Abstract: There has been considerable interest in using decomposition methods in epidemiology (mediation analysis) and economics (Oaxaca-Blinder decomposition) to understand how health disparities arise and how they might change upon intervention. It has not been clear when estimates from the Oaxaca-Blinder decomposition can be interpreted causally because its implementation does not explicitly address potential confounding of target variables. While mediation analysis does explicitly adjust for confounders of target variables, it typically does so in a way that effectively entails equalizing confounders across racial groups, which may not reflect the intended intervention. Revisiting prior analyses in the National Longitudinal Survey of Youth on disparities in wages, unemployment, incarceration, and overall health with test scores, taken as a proxy for educational attainment, as a target intervention, we propose and demonstrate a novel decomposition that controls for confounders of test scores (e.g., measures of childhood socioeconomic status [SES]) while leaving their association with race intact. We compare this decomposition with others that use standardization (to equalize childhood SES [the confounder] alone), mediation analysis (to equalize test scores within levels of childhood SES), and one that equalizes both childhood SES and test scores. We also show how these decompositions, including our novel proposals, are equivalent to implementations of the Oaxaca-Blinder decomposition but provide a more formal causal interpretation for these decompositions.

Keywords: Causal inference, Health disparity, Inequity, Interventional effects, Mediation analysis, Path analysis, Path-specific effect, Oaxaca-Blinder decomposition, Standardization

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From the ^aDepartment of Epidemiology, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD; ^bDepartment of Mental Health, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD; ^cDepartment of Epidemiology, Harvard T.H. Chan School of Public Health, Boston, MA; and ^dDepartment of Biostatistics, Harvard T.H. Chan School of Public Health, Boston, MA.
Data and Code Availability: <https://github.com/jwjackson/SuppMaterials>.
John Jackson was partly funded by the Alexios Smyrle Yerby Fellowship, and Tyler J. VanderWeele was funded by the National Institutes of Health grant ES017876.

The authors report no conflicts of interest.
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www.epidem.com | 825

Prior decompositions (e.g., Kitagawa, O-B) ‘descriptive’

Causal mediation challenges

Challenges with non-manipulable exposures

How much would gaps change if we intervened on a target?

Many recent developments Reconciling the descriptive framework of KBO with causal inference.

ORIGINAL ARTICLE

Decomposition Analysis to Identify In- for Reducing Dispariti

John W. Jackson,^{a,b,c} and Tyler J. VanderWee

Abstract: There has been considerable interest in using decomposition methods in epidemiology (mediation analysis) and economics (Oaxaca–Blinder decomposition) to understand how health disparities arise and how they might change upon intervention. It has not been clear when estimates from the Oaxaca–Blinder decomposition can be interpreted causally because its implementation does not explicitly address potential confounding of target variables. While mediation analysis does explicitly adjust for confounders of target variables, it typically does so in a way that effectively entails equalizing confounders across racial groups, which may not reflect the intended intervention. Revisiting prior analyses in the National Longitudinal Survey of Youth on disparities in wages, unemployment, incarceration, and overall health with test scores, taken as a proxy for educational attainment, as a target intervention, we propose and demonstrate a novel decomposition that controls for confounders of test scores (e.g., measures of childhood socioeconomic status [SES]) while leaving their association with race intact. We compare this decomposition with others that use standardization (to equalize childhood SES [the confounders] alone), mediation analysis (to equalize test scores within levels of childhood SES), and one that equalizes both childhood SES and test scores. We also show how these decompositions, including our novel proposals, are equivalent to implementations of the Oaxaca–Blinder decomposition but provide a more formal causal interpretation for these decompositions.

Keywords: Causal inference, Health disparity, Inequity, Interventional effects, Mediation analysis, Path analysis, Path-specific effect, Oaxaca Blinder decomposition, Standardization

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ORIGINAL ARTICLE

Meaningful Causal Decompo Health Equity Research

Definition, Identification, and Estimato
Weighting Framework

John W. Jackson^{a,b,c,d,e}

Abstract: Causal decomposition analyses can help build the evidence base for interventions that address health disparities (inequities). They ask how disparities in outcomes may change under hypothetical intervention. Through study design and assumptions, they can rule out alternate explanations such as confounding, selection bias, and measurement error, thereby identifying potential targets for intervention. Unfortunately, the literature on causal decomposition analysis and related methods have largely ignored equity concerns that actual interventions would respect, limiting their relevance and practical value. This article addresses these concerns by explicitly considering what covariates the outcome disparity and hypothetical intervention adjust for (so-called allowable covariates) and the equity value judgments these choices convey, drawing from the bioethics, biostatistics, epidemiology, and health services research literatures. From this discussion, we generalize decomposition estimands and formulae to incorporate allowable covariate sets (and thereby reflect equity choices) while still allowing for adjustment of non-allowable covariates needed to satisfy causal assumptions. For these general formulae, we provide weighting-based estimators based on adaptations of ratio-of-mediator-probability and inverse-odds-ratio weighting. We discuss when these estimators reduce to already used estimators under certain equity value judgments, and a novel adaptation under other judgments.

Keywords: Allowability; Causal inference; Disparity; Equity; Ethics; Interventional effects; Mediation analysis; Oaxaca–Blinder decomposition

Editors' note: Related article appears on page 291.

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Article

The Gap-Closing Estimand: A Causal Approach to Studying Interventions That Disparities Across Categories

BY ANG Y

Department of Sociology, University

We introduce a new nonparametric causal estimand that identifies the mechanisms group-based outcome disparities; group differences in effects, and (3) selection effects. Our approach refines compositions in causal analysis by explaining isolates conceptually distinct decomposition. In uniquely identifies different generating mechanism. Our causal explanation of disparity to change disparities. We present composition, where the latent within levels of certain covariates are $\sqrt{n}$ -consistent, asymptotically multiply robust. We apply college graduation causally (the disparity in adult income parents). Empirical played by the new selection.

Ian Lundberg¹

Abstract

Disparities across race, gender, and research. But rather than only inform interventions to close the how much a gap (e.g., income equalize a treatment (e.g., access to analyses, this type of rose gap-closing estimands place categorical making them play the role of non-manipulable empower researchers to study learning estimators designed for can directly inform policy: if we changed treatment assignments comes? I provide open-source support these methods.

1. Introduction. Social and health parity between groups in terms of time. For example, how much, and in what racial disparities in health (Howe and gender wage gap (Cha, Weeden and in the association between parents' at Muller and Ridge (1995)? The community the mechanisms by which a treatment disparity.

Prior research has addressed successfully appropriate for the task of causal Kitagawa–Blinder–Oaxaca (KBO) decomposition (1973)) are defined in terms of a causal question by design (Fortin, Le Stewart (2021), p. 542). Second, although formulated in causal terms, that observed group disparities. Thir (VanderWeele and Robinson (2024)) conflate distinct interpretability, explanatory power, and

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Key words and phrases: Social inequality,

Check for updates

Original Article

The Target Study A Conceptual Framework for Disparity

John W. Jackson^{1,2,3},
Romsai T. Boonyasai^{5,1}

Abstract

We present a conceptual model where social groups may be sociocultural. Our model, based on random sample selection, reports disparity or to assess disparity variables that eliminates forms differential timing of enrollment

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Data Availability Statement included at

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Original Article

ROYAL STATISTICAL SOCIETY
Series A
Statistics in Society

Disparity analysis: a tale of two approaches

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Abstract

To understand patterns of social inequality, social science research has typically relied on statistical models linking the conditional mean of an outcome variable to a set of explanatory factors. A prime example of this approach is the Kitagawa–Oaxaca–Blinder (KOB) method. By fitting two linear models separately for an advantaged group and a disadvantaged group, the KOB method decomposes the between-group outcome disparity into two parts: a part explained by group differences in background characteristics, and an unexplained part often dubbed ‘residual inequality’. In this article, we explicate, contrast, and extend two distinct approaches to studying group disparities, which we term the descriptive approach, as epitomized by the KOB method and its variants, and the prescriptive approach, which focuses on how a disparity of interest would change under a hypothetical intervention to one or more manipulable treatments. For the descriptive approach, we propose a generalized nonparametric KOB decomposition that considers multiple explanatory variables sequentially. For the prescriptive approach, we introduce a variety of stylized interventions, such as lottery-type and affirmative-action-type interventions that close between-group gaps in treatment. We illustrate the two approaches to disparity analysis through an application to the Black–White gap in college completion rates in the U.S.

Keywords: causal inference, college completion gap, debiased machine learning, disparity analysis, interventional disparities, Kitagawa–Oaxaca–Blinder (KOB) decomposition

1 Introduction

Social and economic inequalities are a perennial topic of economic, demographic, and sociological research. To understand the patterns and trends of various forms of inequality, quantitative social science research has typically relied on statistical models that link the conditional mean (and, sometimes, other distributional statistics such as the conditional variance) of an outcome of interest, such as logged hourly wage, to a range of explanatory variables, such as education (e.g. Juhn et al., 1993; Katz and Murphy, 1992), occupation (e.g. Mow and Kalleberg, 2010), and union membership (e.g. DiNardo et al., 1996; Western and Rosenfeld, 2011). A prime example of this approach is the Kitagawa–Oaxaca–Blinder (KOB) decomposition method (Blinder, 1973; Kitagawa, 1955; Oaxaca, 1973; see also Althausser and Wigler, 1972; Duncan, 1968).¹ By fitting two linear regression models, one for an advantaged group (e.g. Whites) and one for a disadvantaged group (e.g. African Americans), it decomposes the between-group disparity in the mean outcome into two parts: a part accounted for by group differences in a set of explanatory variables and an unexplained part often dubbed ‘residual inequality’. While the canonical KOB method decomposes the gap in mean outcomes based on linear models, various extensions of the KOB method have been proposed, including decomposition based on nonlinear outcome models (Bauer and

¹ For a long time, the KOB method was (and is still by many) called the Oaxaca–Blinder or Blinder–Oaxaca decomposition (Blinder, 1973; Oaxaca, 1973). In recent years, it has been increasingly referred to as the Kitagawa–Oaxaca–Blinder or Kitagawa–Blinder–Oaxaca decomposition, in recognition of Kitagawa (1955), who proposed the same method in a nonparametric framework. We follow this emerging convention and use the KOB acronym.

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See Jackson and VanderWeele (2018); Jackson (2021); Lundberg (2024); Yu and Elwert (2025); Jackson et al. (2025); Opacic, Wei, and Zhou (2026)

Youth accident deaths: what would we intervene on?

Danish-origin adolescents die in accidents more often than immigrant-origin peers (IRR 1.7 vs first-generation)

Alcohol is a plausible pathway: 16% of fatal accidents; Danish-origin has 4x the odds of an alcohol-related death

But immigrant-origin families are poorer, so adjusting for income *widens* the gap (IRR 1.7 → 2.6)

What if Danish-origin youth drank like their immigrant-origin peers?

Udesen et al. (2023); Kruckow and Tolstrup (2024) (ethnic IRRs inverted from Kruckow supplementary table 2)

bmjmedicine



► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/bmjmed-2023-000685>).

National Institute of Public Health, University of Southern Denmark, Copenhagen, Denmark

All cause and cause specific mortality in 15-24-year-olds in Denmark 2010 to 2022: nationwide study of socioeconomic predictors

Sofie Kruckow , Janne S Tolstrup

ABSTRACT

OBJECTIVE To assess inequalities in all cause and cause specific mortality in young people and if there are differences across gender and age groups.

DESIGN Nationwide cohort study of socioeconomic predictors.

SETTING Denmark, 1 January 2010 to 31 December 2022

association showing a higher mortality with lower levels of socioeconomic position. For instance, incidence rate ratios of all cause mortality related to parents' education was 2.3 (95% CI 2.0 to 2.7) for elementary level, 1.5 (1.3 to 1.6) for low, and 1.3 (1.1 to 1.4) for medium level as compared with high level. For deaths, incidence rate ratios of elementary education level compared with the most

Alcohol-related mortality in 15-24-year-olds in Denmark (2010-2019): a nationwide exploratory study of circumstances and socioeconomic predictors



Caroline Holt Udesen, Signe Skovgaard Hviid, Ulrik Becker, and Janne S. Tolstrup*

National Institute of Public Health, University of Southern Denmark, Studiestraede 6, DK-1455 Copenhagen, Denmark

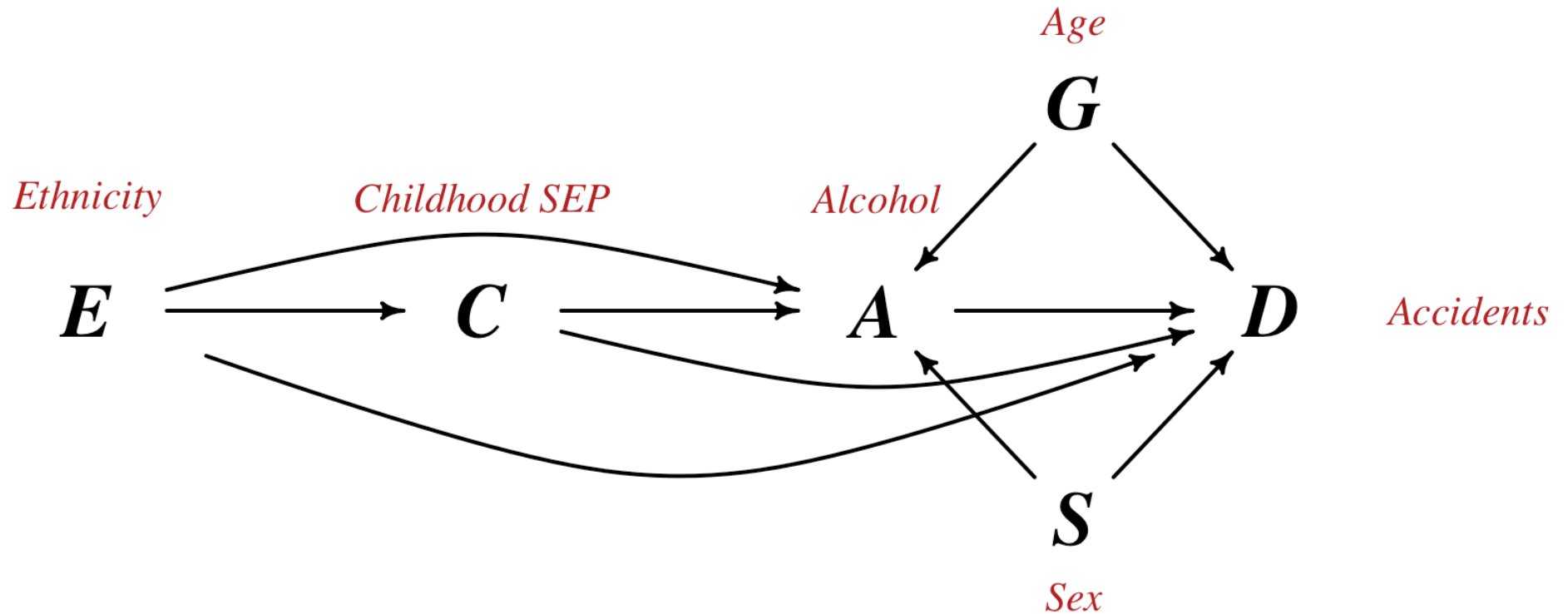
Summary

Background Adolescents and young adults aged 15-24 years are disproportionately affected by unnatural deaths, including accidents, suicide and interpersonal violence for which alcohol is a leading risk factor. We aimed to explore the extent of and circumstances surrounding alcohol-related deaths in young people aged 15-24 years and whether socioeconomic background and ethnicity differ in those who died due to alcohol-related causes as compared to the background population.

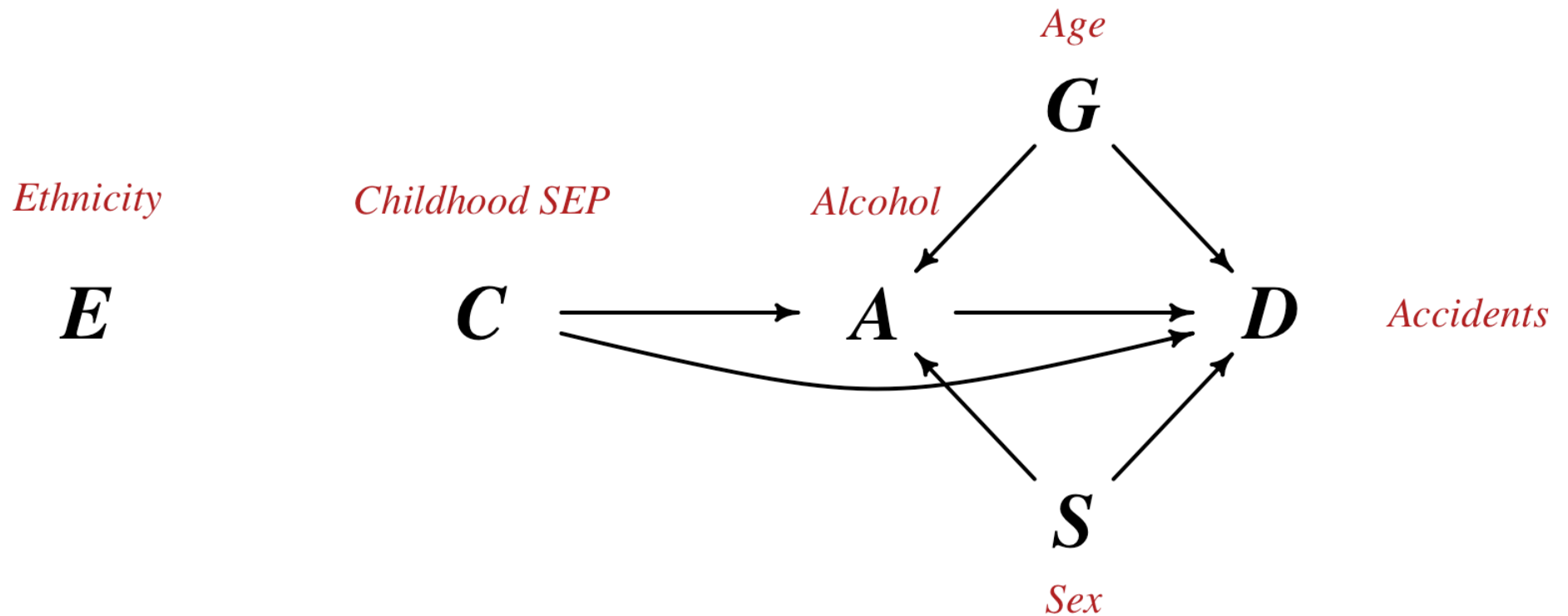


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Conceptualizing the DAG (assumptions)



Different from causal mediation (still valid)



Steps for
pursuing
interventional
decomposition

1 Define the gap and intervention

Consider feasibility, implementation

2 Draw the DAG

Which covariates to adjust for (allowability)

3 Defend the assumptions

Exchangeability; positivity; consistency

4 Estimate the model

Outcome model, weighting, doubly robust

5 Compare factual and counterfactual

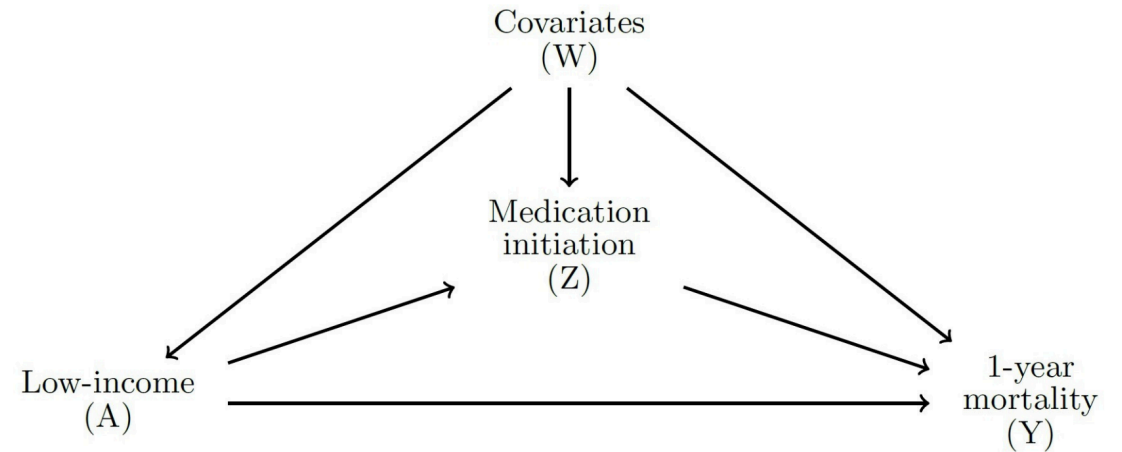
Consider effect scale, uncertainty

Recent example

Theory and methods

Estimating effects of targeted public health interventions using the interventional disparity indirect effect among the exposed

Amalie Lykkemark Møller ^{1,2,3} Kathrine Kold Sørensen,⁴ Nerissa Nance,^{5,6} Julie Thietje Mortensen,⁵ Thomas Alexander Gerds,² Christian Torp-Pedersen,^{2,4} Helene Charlotte Wiese Rytgaard²



we estimate the expected 1-year mortality among the low-income patients in a hypothetical world where they initiated medication as often as high-income patients...

Moller results

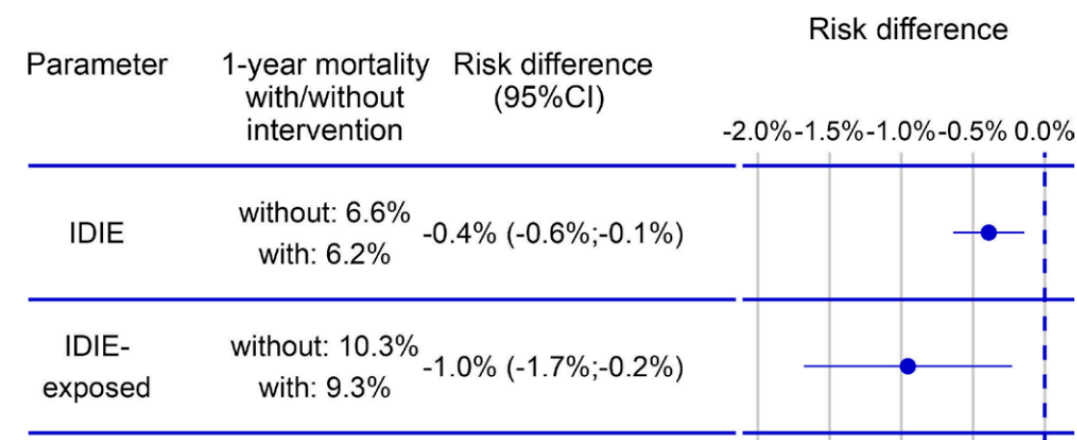


Figure 3 Comparison of the interventional disparity indirect effect (IDIE) and the interventional disparity indirect effect among the exposed (IDIE-exposed) in the study of medication initiation among heart failure patients. We studied a potential barrier-removing intervention under which low-income heart failure patients become as likely as similar high-income patients to initiate treatment. Patients with missing information on the educational level were excluded (n=50). IDIE, interventional disparity indirect effect; IDIE-exposed, interventional disparity indirect effect among the exposed.

IDIE-exposed is the expected reduction in 1-year mortality among low-income heart failure patients in a hypothetical world where they initiated medication with the same probability as observed among similar high-income patients, rather than their observed probability.

Summary: Interventional decomposition

When to use

Counterfactual inequality under a hypothetical intervention on a mediator

Extensions

Gap-closing estimand; weighting framework; nonparametric estimation

Caveats

Causal assumptions; well-defined intervention; choices about ‘allowable’ covariates

Summary

Various decomposition techniques exist that may be useful for analyzing social inequalities in health

Moves beyond measuring to explanations and interventions

Used responsibly, they can help to provide key evidence on why health inequalities exist and change over time.

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Backup

Backup: Kitagawa's three-component version

$$\underbrace{\text{Rate}_A - \text{Rate}_B}_{\text{crude difference}} = \overbrace{\sum_i R_{Bi}(P_{Ai} - P_{Bi})}^{\text{composition}} + \overbrace{\sum_i P_{Bi}(R_{Ai} - R_{Bi})}^{\text{rate}} + \overbrace{\sum_i (R_{Ai} - R_{Bi})(P_{Ai} - P_{Bi})}^{\text{interaction}}$$

Composition				Rate				Interaction			
Group	R_{Bi}	$P_{Ai} - P_{Bi}$	Contrib.	Group	P_{Bi}	$R_{Ai} - R_{Bi}$	Contrib.	Group	$R_{Ai} - R_{Bi}$	$P_{Ai} - P_{Bi}$	Contrib.
Young	20	-0.20	-4.0	Young	0.50	-10	-5.0	Young	-10	-0.20	+2.0
Old	30	+0.20	+6.0	Old	0.50	+10	+5.0	Old	+10	+0.20	+2.0
Sum			+2.0	Sum			0.0	Sum			+4.0

Total = +2.0 + 0.0 + 4.0 = **+6.0** = crude gap (31 – 25)

Add the interaction to composition: +6.0 and 0.0 (A's rates). Add it to rate: +2.0 and +4.0 (B's rates).

Overall
contributions
are *net*

Contribution by Gestational Age, Top 3 vs. Denmark and Austria
Same decomposition, broken out by GA stratum instead of summed

